The Brazilian Geodetic System — Part 2: the vertical network

From sea level to REALT: the zero, the leveling, the adjustments and the two datums — four classes

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Leveling (least squares) and Coordinate Systems Simulators

What we will see

1. Where we left off
2. The sea as a ruler
3. Brazil's zeros
4. The zero is not the geoid
5. Wrap-up
6. Class 2: Leg by leg
7. Class 3: The adjustments and REALT
8. Class 4: The zero, now and next

How to use

Four 50-minute classes. Each *Your turn* has the answer on the next click. Formulas and complete computations are in the **appendix**.

This is Part 2

Part 1 (the previous series)

Position: SIRGAS2000, how Brazil knows *where* each point is — latitude, longitude and height *above the ellipsoid*.

Part 2 (this series)

The **altitude**: the distance *above the sea*. Where the zero is, how the zero is carried to your door, and why the same marker has had several altitudes.

Part 2 also **starts from scratch**. And it **does not repeat** the geoid classes: what they already taught is only recalled.

Four classes

Class 1 · Where is the zero?

Mean sea level, the tide gauge and Brazil's *zeros*: Torres, Imbituba and Santana.

Class 2 · Leg by leg

Geometric leveling, the error that grows with $\sqrt{K}$, and the network of benchmarks.

Class 3 · Adjustments, REALT

Why one benchmark has had several altitudes; the adjustments up to REALT-2018.

Class 4 · The zero, now and next

Tide gauges with GNSS, the two datums that disagree, and practical use.

Each class is 50 minutes long, starts by recalling the previous one and ends with a summary. We will go **slowly**: one idea per slide.

Class 1

Where is the zero?

class 1 of 4

Where we left off

Today's questions

1. Why does altitude need a "zero", and who chooses where it is?
2. How is sea level measured — and for how long?
3. What has been Brazil's zero, and why are there two today?
4. Is that zero really "the sea"?

Deal

By the end of the class you will be able to answer all four — in your own words.

They come back on the last slide.

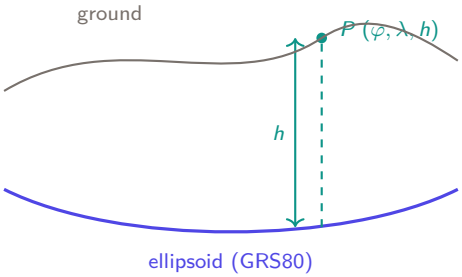
What Part 1 left us

Part 1 ended with a **complete position**, measured by satellite in SIRGAS2000:

1. the **latitude** φ;
2. the **longitude** λ;
3. the **height** h — *above the ellipsoid*.

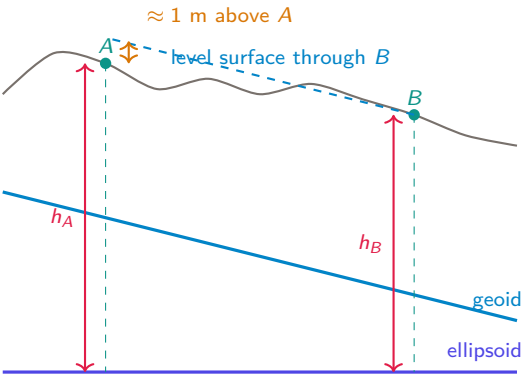
Something is missing

The ellipsoid is a *mathematical* surface. Whoever builds a road, a dam or a sewer wants to know **where the water flows**.



h is measured along the normal to the ellipsoid.

Reminder: water does not flow along the ellipsoid



	A	B
h (m)	12.0	10.0
N (m)	6.0	3.0
$H = h - N$ (m)	6.0	7.0

Looking only at h, A seems *higher*. But water flows from B to A: B is higher in H.

Seen in the geoid classes. Figure not to scale; didactic numbers.

What we have seen, and what is new here

Already seen in the geoid classes

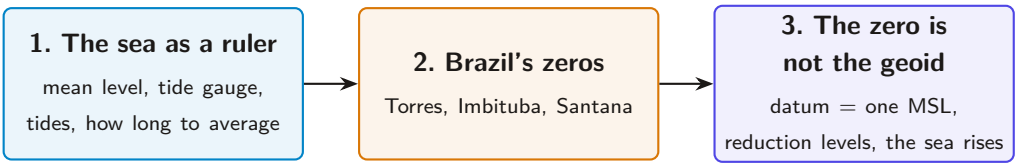
- geoid and level surfaces;
- geopotential number C and kinds of height;
- normal height $H^* = C/\bar{\gamma}$, quasi-geoid;
- IHRS and IHRF; the hgeoHNOR2020 model.

Here, only *recalled*.

New in Part 2

- the **sea** and the **tide gauge**: mean sea level;
- the **vertical datum**: a chosen zero;
- **leveling** and the error that grows with $\sqrt{K}$;
- the **network** of benchmarks;
- the **adjustments** up to **REALT-2018**;
- Brazil's **two zeros**.

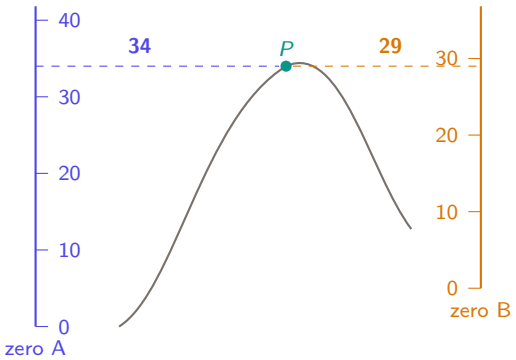
The thread of the class



At the end, a bridge to **Class 2**: how to carry the zero from the coast to your door.

The sea as a ruler

A ruler needs a zero



Altitude is the reading of a vertical "ruler" that starts at a **zero**.

The same point P reads **34** on one ruler and **29** on another, only because the zero was put somewhere else.

Moral

Changing the zero adds the *same number* to every reading. For a whole country, everyone must use **the same zero**. That zero has a name: **vertical datum**.

The sea never stands still

What moves the water	Scale	Typical effect
waves	seconds	up and down all the time
tides	hours	two highs a day
wind and pressure	days	piles up or pushes the water away
seasons	months	warm water takes more volume
Moon's nodes	18.6 years	changes the tide a little

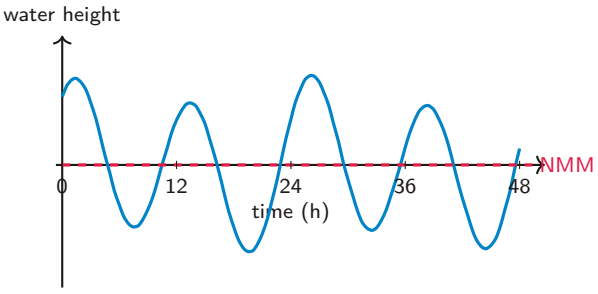
Which is "the" sea level?

If you measure in the morning, in the afternoon, at full moon or in a storm, the number changes. There is no single height of the sea.

Way out

Take the **average**: let what goes up and down cancel out.

Mean sea level



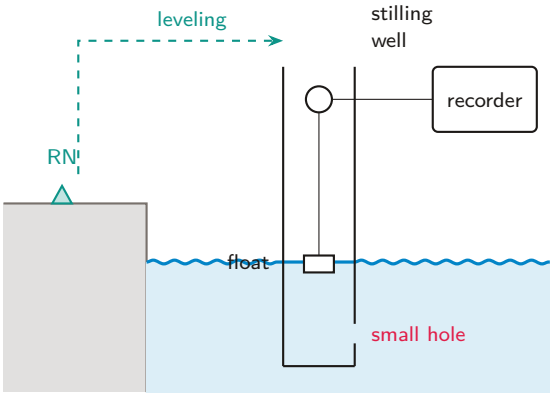
Illustrative two-day curve (not a real port).

The water height is recorded **every hour**, for a long time, and averaged:

$$\mathrm{NMM} = \frac{1}{n}\sum_{i=1}^{n} h_i$$

The average erases waves, tides and wind. What remains is the **mean sea level** (*MSL*).

How it is measured: the tide gauge



- The pipe has a **small hole**: the water enters slowly and the level inside goes up and down *without the waves*.
- A **float** follows the level; a wire carries the motion to the **recorder** (once on paper, now in a file).
- The instrument's zero is tied to a **benchmark** (RN) on land, by **leveling**.

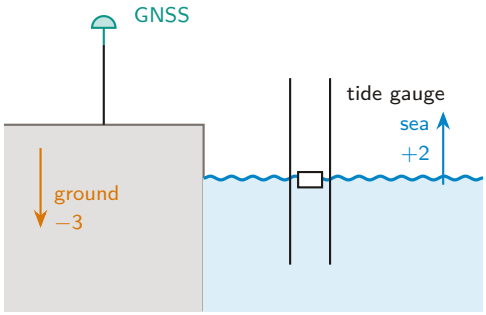
Today's tide gauges

Type	How it measures	Strength	Care
Tide staff	someone reads the water height	simple, cheap	depends on the observer
Float in a well	float, wire, recorder	filters the waves	the hole can clog
Pressure	bottom sensor weighs the water column	easy to install	correct for air pressure, density
Radar	measures the distance to the surface	nothing in the water	waves, heavy rain

In geodesy

A geodetic station adds weather sensors and a **GNSS antenna** to the level sensor. The next idea explains why.

The ground moves too



mm/year (didactic numbers)

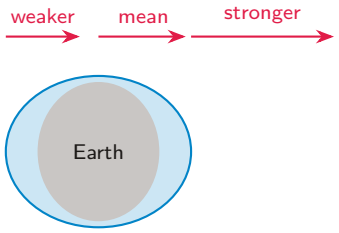
The tide gauge is *fixed to the ground*. It measures the sea **relative to the ground**.

If the ground sinks, the gauge "sees" the sea rise *more*: in the example the sea rises 2 and the ground drops 3, so the instrument records +5 mm/year.

That is why GNSS

A fixed GNSS antenna next to it measures the vertical motion of the ground. That separates **what is sea** from what is **ground**.

Why the Moon rules the tides



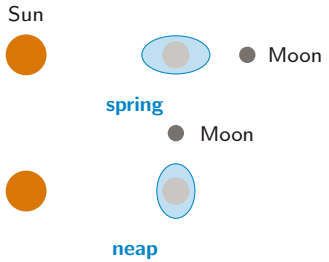
the difference stretches the water on *both* sides

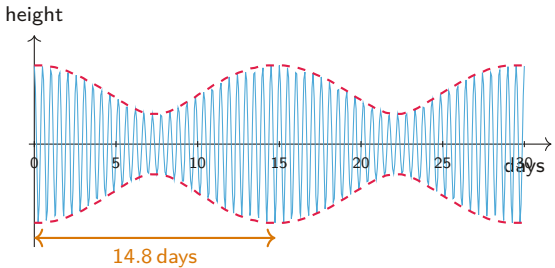
The Moon pulls the water on this side slightly *more* than it pulls the Earth, and the water on the far side slightly *less*. That **difference** stretches the ocean into **two highs**: one under the Moon, one on the opposite side.

The Earth turns under these two highs; since the Moon also moves along its orbit, the interval between two high waters is 12.42 **h** (not 12).

The Sun does the same, but weaker (about 46 % of the Moon).

Spring and neap tides





With Sun and Moon **aligned** (new or full moon), the two forces add up: **spring** tide. At **right angles** (quarters), one hinders the other: **neap** tide. The back-and-forth repeats every 14.8 **days**.

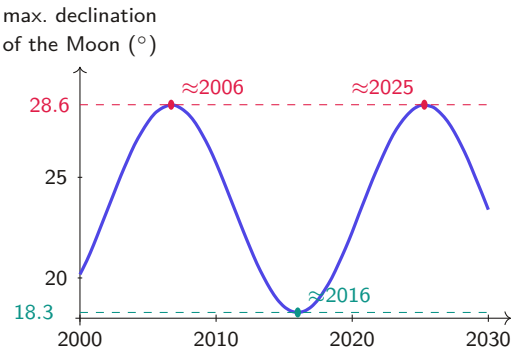
Longer rhythms

Period	What drives it	What changes
12.42 h	the Moon goes around the Earth	two high waters a day
14.8 days	Sun and Moon aligned or at right angles	spring and neap tides
1 year	seasons: heat, wind, rain	summer MSL is not winter MSL
18.6 years	the Moon's orbit "turns" in space	changes the tide and MSL a little

Moral

One month of data only catches the short rhythm. For the *real* mean level you need to observe a **whole** cycle — the longest is the **18.6-year** one.

The 18.6-year cycle



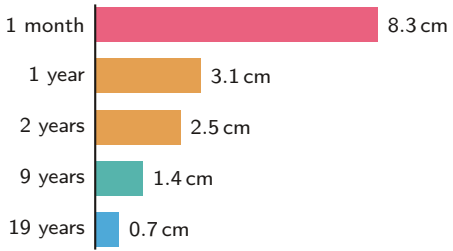
The Moon's orbit is tilted 5.1° relative to the plane of the Earth's orbit. That tilt "points" to one side and slowly turns: one full turn every 18.6 **years**.

That is why the Moon's *maximum declination* (how far it strays from the celestial equator) swings between 18.3° and 28.6°.

You have seen this cycle

It is the same period as the main **nutation**, in the series on celestial reference systems.

How long to average?



Typical error of the estimated mean level, by observation time.

Illustrative simulation (synthetic series, not measured).

- **1 month**: error of several cm.
- **1 year**: still a few cm.
- **9 years** (almost a nodal cycle): about 1 cm.
- **19 years**: a few mm.

Practice

Tide services use **19-year** series (e.g., NOAA's datum epoch): the nodal cycle in whole years.

Your turn

The rhythm of the tide

(a) Today the high tide was at 06:10. About what time should the next one be? And tomorrow's first?

(b) In 18.6 years, roughly how many high tides occur? (Use an interval of 12.42 h between them.)

Your turn

The rhythm of the tide

(a) Today the high tide was at 06:10. About what time should the next one be? And tomorrow's first?

(b) In 18.6 years, roughly how many high tides occur? (Use an interval of 12.42 h between them.)

Answers

(a) $12.42\,\mathrm{h} = 12\,\mathrm{h}\,25\,\mathrm{min}$: the next high is at **18:35**. Each day's high comes about 50 min later: tomorrow's first is near **07:00**.

(b) $\dfrac{18.6 \times 365.25 \times 24}{12.42} \approx 13\,100$.

That is a *lot* of tides: each one is a data point of the level.

Brazil's zeros

The first zero: Torres (1919–1920)

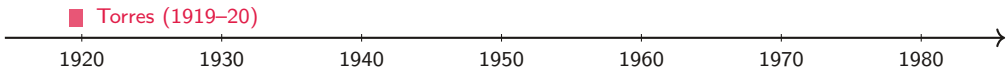
Before there was a leveling network, there already was a **provisional** zero.

The Comissão da Carta Geral do Brasil installed a tide gauge at **Torres** (RS). The mean level from **February 1919 to February 1920** — about **one year** — became the zero.

It was provisional on purpose. Comparisons with the Laguna and Florianópolis gauges led to the suspicion of an error of **a decimeter or more**.

Remember the bars?

With *one year* of data, the simulation in this class gave a typical error of a few centimeters. The historical suspicion (10 cm or more) is of the *same order*.



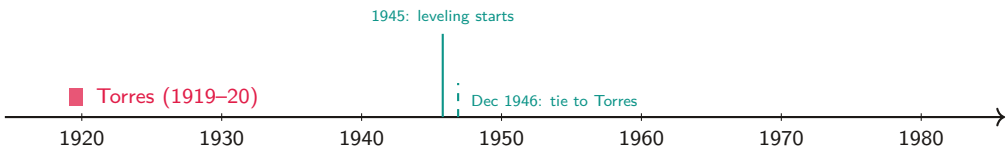
1945: precision leveling begins

- On **13 October 1945**, the Leveling Section of the **Conselho Nacional de Geografia** (CNG, of IBGE) starts precision leveling (date per H. Beserra, in the *RBG*).
- The network is born at **Urussanga** (SC) and grows *along the roads*.
- Only in **December 1946** is the network tied to the Torres gauge: for the first 14 months the heights still had no *zero*.

A question

Torres is on the coast of Rio Grande do Sul. How does the zero reach Urussanga, *hundreds of kilometers* away, with no sea on the way?

By leveling: that is Class 2.

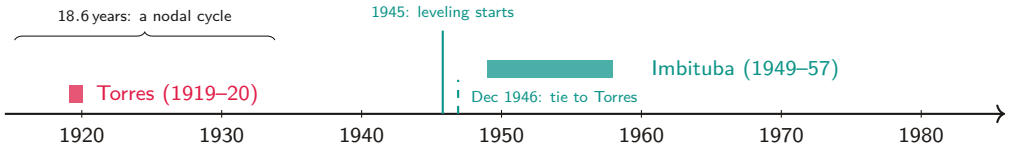


Imbituba: a new tide gauge (1949–1957)

- At the **port of Imbituba** (SC), there are tide gauge records since 1948 (the PSMSL series run from 1948 to 1968).
- The **datum** was defined with the observations of **1949 to 1957**: **9 years**.
- In REALT-2018, **RN 4X** is the reference of the Imbituba datum.

Nine years is almost half the cycle

$9/18.6 \approx 0.48$: the window covers almost *half* of the nodal cycle. It is much better than Torres' one year — but not today's 19-year standard.



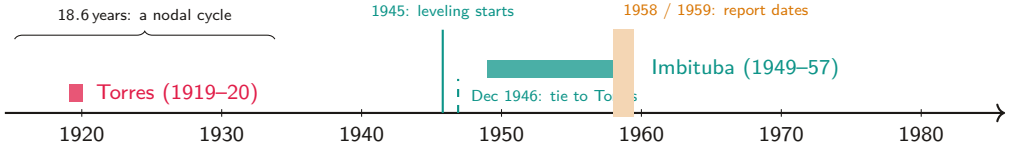
Torres and Imbituba: two documented dates

The **2011 IBGE report**, p. 8, dates the replacement of Torres by Imbituba to **1958**, with about 30 000 km of lines.

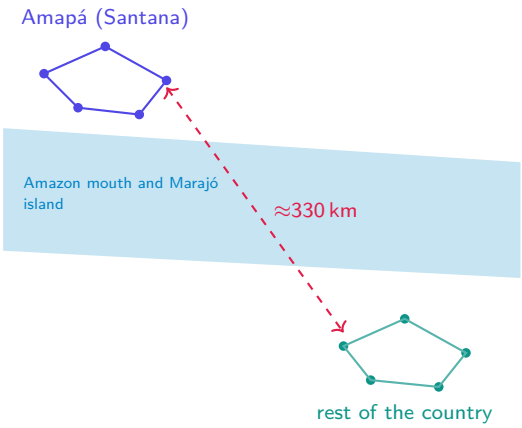
The **REALT report, 2nd ed. (2019)**, p. 13, describes Imbituba's adoption in **1959**, when the third adjustment ended. The 1948 and 1952 adjustments used Torres; those from 1959 onward used Imbituba.

A historical qualification

The two official publications give different years. The timeline records both without assuming they describe exactly the same administrative act.



The Amazon cannot be crossed



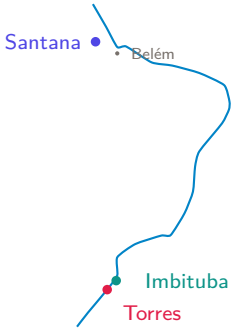
High-precision geometric leveling walks *on land*, with level and rod. It **cannot cross** the Amazon mouth and Marajó island.

That is why the small **Amapá** network was never tied to that of the rest of the country. It got its **own zero**: mean sea level at the **port of Santana** in **1957–58**, adopted in **1980**. The old benchmarks were established in **1980–81**.

IBGE (2019), pp. 13 and 44.

The nearest benchmarks of the two networks are about 330 km apart.

Brazil's two zeros



sketch: coast linking cities

	Imbituba (SC)	**Santana** (AP)
MSL observed	**1949–1957** (9 years)	**1957–1958** (2 years)
Datum RN	**4X**	**9329T**
Where it applies	almost the whole country	Amapá
Adopted	1958 / 1959[a]	1980

[a] IBGE (2011), p. 8: 1958; IBGE (2019), p. 13: 1959. The heights rest on two MSL values; their relation returns in Class 4.

Your turn

Three windows, three zeros

(a) What fraction of the nodal cycle (18.6 years) does each window cover: Torres (1 year), Santana (2 years), Imbituba (9 years)?

(b) In the simulation the MSL error is 3.1 cm with 1 year and 1.4 cm with 9 years. How many times larger would Torres' error be than Imbituba's?

Your turn

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Answers

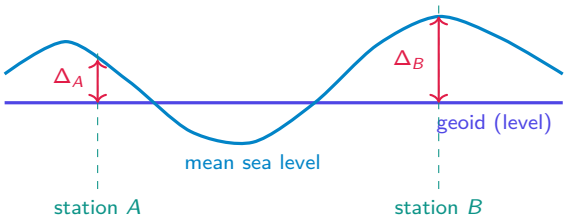
(a) $1/18.6 \approx 0.05$; $2/18.6 \approx 0.11$; $9/18.6 \approx 0.48$.

(b) $3.1/1.4 \approx 2.2$ times.

Careful: the bars come from a simulation. What matters is the *idea*: the more of the cycle covered, the more stable the zero.

The zero is not the geoid

Mean sea level is not a level surface



Exaggerated sketch, not to scale.

The **geoid** is a level surface: *still* water would sit on it.

The real sea does not stand still. **Currents, winds, pressure, temperature and salinity** push it up or down by tenths of a meter.

Consequence

The MSL of each station lies a little *above* or *below* the geoid, and the difference $(\Delta_A,\ \Delta_B)$ **changes from place to place**.

Vertical datum: one value, at one station, at one epoch

WHERE

one station: the Imbituba tide gauge (or Santana's)

WHEN

a window of years: 1949–1957 (Santana: 1957–1958)

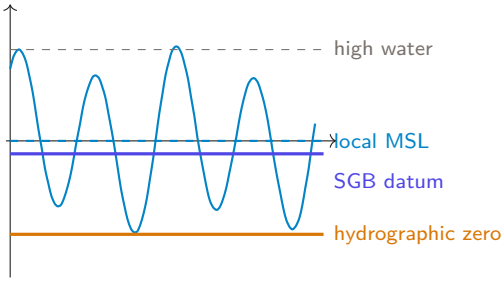
HOW

the mean sea level in that window *equals zero altitude*, by convention

Definition

A **vertical datum** is *one value of mean sea level*, measured *at one station* during *one time window*, to which zero altitude is assigned. The rest of the country has no tide gauge: the zero reaches it by **leveling** — Class 2.

Other zeros: reduction level and hydrographic zero



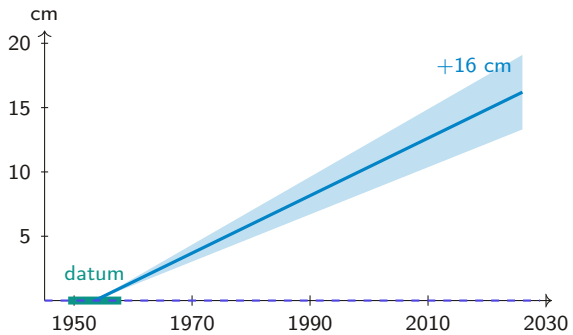
Nautical chart (Navy): depth is counted from the **reduction level** (hydrographic zero), a plane near the *lowest* tide. This way the navigator never finds less water than the chart says.

Altitude (IBGE): counted from the SGB **datum**, a *mean* level.

They are different zeros

In each port the relation between them is different, and IBGE studies and publishes it (RMPG network, Class 4). The IHO recommends, for charts, the lowest astronomical tide, computed with ≥ 18.6 years.

The sea rises — and the altitude does not change



MSL relative to the Imbituba datum zero, with the rate 2.24 ± 0.4 mm/year (UFPR study, 2017).

The datum uses the 1949–57 window. If the study's rate were constant, the projection to 2026 would be about 16 **cm**. This is an illustrative extrapolation.

The **altitude of a benchmark** does not change because of it: it is "so many meters above the mean level of *1949–57*", and it still is.

Same study: the gauge's GNSS indicated subsidence of about 3 mm/year (short campaigns). Values from one study, not official.

Your turn

Today's sea

(a) MSL rises 2.24 mm/year. From 1953.5 (middle of the window) to 2026, 72.5 years pass. How many centimeters has it risen?

(b) A benchmark on the coast has altitude 7.000 m in the datum. How high is it above *today's* sea? (Ignore tides, waves and ground motion.)

Your turn

Today's sea

(a) MSL rises 2.24 mm/year. From 1953.5 (middle of the window) to 2026, 72.5 years pass. How many centimeters has it risen?

(b) A benchmark on the coast has altitude 7.000 m in the datum. How high is it above *today's* sea? (Ignore tides, waves and ground motion.)

Answers

(a) $2.24 \times 72.5 = 162\,\text{mm} \approx \mathbf{16\,cm}$ (± 3 cm from the rate's uncertainty).

(b) $7.000 - 0.162 \approx \mathbf{6.84\,m}$.

The official altitude is *the same*; only the distance to today's sea changed.

Stop and think

A tempting idea

"If the sea rises, just *update the zero* every year. The zero then follows the sea."

Why does Brazil *not* do that?

Stop and think

A tempting idea

"If the sea rises, just *update the zero* every year. The zero then follows the sea."

Why does Brazil *not* do that?

Because

- **all** altitudes (maps, projects, laws) depend on the zero;
- the datum is a *convention* fixed at an epoch;
- what is done is to **measure the difference** between the datum and today's sea (RMPG network, Class 4) and, in the future, tie everything to a *global* zero.

Wrap-up

The four questions, answered

1. **Why does altitude need a zero?** Because it is the reading of a ruler, and every ruler starts somewhere. The zero is a **convention**: the mean sea level at a station, at an epoch.

2. **How is sea level measured — and for how long?** With a **tide gauge**, averaging many readings. One year leaves a few centimeters of error; the standard is **19 years**, the Moon's nodal cycle.

3. **What has been Brazil's zero, and why are there two today?** Torres (1919–20, provisional), then **Imbituba** (1949–57, adopted in 1958–59). The Amazon cannot be crossed: Amapá uses **Santana** (1957–58).

4. **Is that zero really "the sea"?** No: it is the sea *of that window*. The sea rises, local MSL is not a level surface, and the nautical charts' zero is another one.

Take home

Four ideas

1. Altitude is the reading of a ruler; the zero is a **choice**.
2. Mean sea level = **average over time**; the standard is 19 years.
3. Datum = **one value, at one station, at one epoch**; Brazil has two.
4. The sea rises; the benchmark's altitude **does not change**.

Bridge to Class 2

The zero is at Imbituba. But your construction site is *hundreds of kilometers* from the sea. **How is the zero carried there?**

Answer: with a level, a rod and lots of patience — **leveling**.

Class 2: Leg by leg

Today's questions

1. How is the zero carried from the sea to a point hundreds of kilometers away?
2. Where does leveling error come from — and why does it grow with the square root of the distance?
3. What does the network of benchmarks that Brazil built look like?
4. Why does a loop measured *without error* not close at zero?

Deal

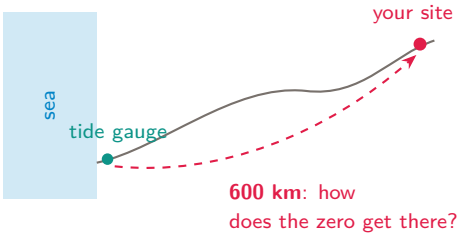
By the end of the class you will be able to answer all four — in your own words.

They come back on the last slide.

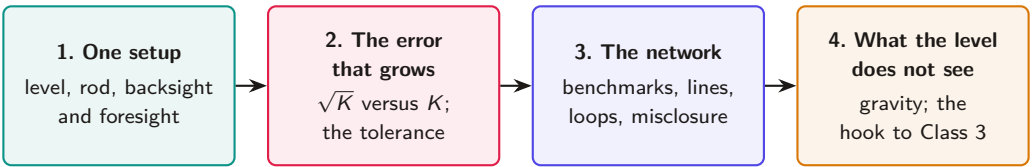
What the previous class left us

In Class 1, three ideas:

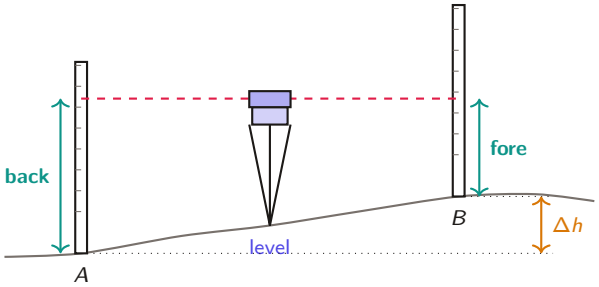
1. altitude is the reading of a ruler; the **zero** is a choice;
2. Brazil's zero is the **mean sea level** at a tide gauge, in a window of years: Imbituba (1949–57) and Santana (1957–58);
3. the sea rises, but the altitude of a benchmark **does not change**.



The thread of the class



The level and the rod



The **level** is a telescope that, on its tripod, always stays **horizontal**: the line of sight draws a plane of constant height.

On the **rods** we read how far above the ground that line passes: **backsight** (to the point of known altitude) and **foresight** (to the new point).

Height difference

$$\Delta h = \text{back} - \text{fore}$$

One setup, step by step

1. Sight the rod at A (altitude 12.000 m): **backsight** $= 1.482$ m.
2. Turn to the rod at B: **foresight** $= 0.937$ m.
3. $\Delta h = 1.482 - 0.937 = +0.545$ m.
4. $H_B = 12.000 + 0.545 = 12.545$ m.

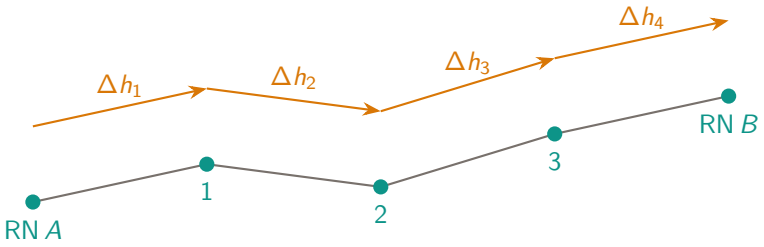
Why back − fore?

The line of sight is the same for both readings. The *larger* the rod reading, the *lower* the point. If the backsight (at A) is larger than the foresight (at B), then B is *higher*.

Mind the sign

Swapping backsight and foresight swaps the sign of the height difference. That is why the field book *always* records which is which.

Setup after setup



The zero walks

$$H_B = H_A + \Delta h_1 + \Delta h_2 + \Delta h_3 + \Delta h_4 = H_A + \sum_i \Delta h_i$$

Each setup is a few tens of meters. A leveling line of 1000 km is *thousands* of chained setups — each one carries a small error.

Why short, equal sights?

Three things get in the way, and all three grow with the sight distance:

1. the **curvature of the Earth**: the horizontal line "rises" relative to the ground, $c = s^2/2R$;
2. the **refraction** of the air bends the light ray and eats 13 % of that;
3. the **collimation**: the level's axis may be slightly tilted.

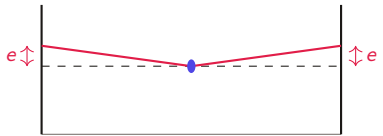
sight	curvature	with refraction
30 m	0.07 mm	0.06 mm
50 m	0.20 mm	0.17 mm
100 m	0.79 mm	0.68 mm

Recipe

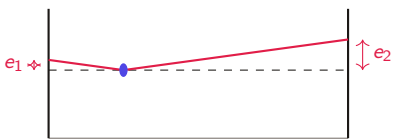
Short sights (small errors) and **equal** ones (what is left cancels): the level goes in the *middle*.

Equal sights: the error cancels

equal distances



unequal distances



An axis tilted by $10''$ (or the curvature) makes both readings come out *too high*, in proportion to the distance.

Equal: e on both; in back − fore the error vanishes.

Unequal: $e_2 - e_1$ is left. With $10''$ and a 20 m difference, that is 0.97 **mm per setup** — over 100 setups, almost **10 cm**.

Forward and back: double leveling

Each **section** between two benchmarks is leveled **twice**: forward (*leveling*) and back (*counter-leveling*).

forward	$+12.347\,\text{m}$
back	$-12.343\,\text{m}$
discrepancy	**4 mm**
mean	$12.345\,\text{m}$

Section with $K = 2.8\,\text{km}$: tolerance $3\sqrt{K} = 5.0\,\text{mm}$. Accepted.

What it is for

- the **discrepancy** is a test: if it is large, something went wrong (a blunder, wind, a badly seated rod) and the section is *redone*;
- the **mean** of the two measurements is better than either;
- errors that change sign on the way back (like some temperature effects) cancel.

High-precision tools

The invar rod

Invar is an iron-nickel alloy that *hardly expands* with heat. The graduation sits on an invar strip stretched along the rod: the length of the "centimeter" does not change with the day's temperature.

The level with a micrometer

A plane-parallel glass plate, turned by a screw, shifts the rod image to "touch" the nearest line: a fraction of a millimeter is read.

The rest is method

- level **always in the middle** (equal sights);
- rod **on a foot plate**, held vertical;
- umbrella over the instrument: the Sun deforms the level;
- hours without strong wind; *forward* and *back* at different times and conditions;
- readings written down to *tenths of a millimeter*.

Try it: the leveling simulator

In the `nivelamento/` simulator, each section between two points receives *Gaussian noise with deviation* $\sigma\sqrt{d}$ — the $\eta\sqrt{K}$ of this class.

1. Generate a network (*Generate New Network*); one point is fixed: the datum.
2. Set σ to $1\,\mathrm{mm}/\sqrt{\mathrm{km}}$ and run (*Run Adjustment*).
3. Double σ: the residuals double.
4. Use *Simulate Gross Error (Blunder)* and watch the test find the bad section.

What to look at

- *Residuals & Data Snooping*: where is the blunder?
- *Global Test* (χ^2): is the network compatible with the given σ?
- *Monte Carlo*: repeat the draw and see the cloud of errors.

The mathematics of the adjustment is in Class 3.

Your turn

One setup

(a) A has altitude 100.000 m. Backsight at A: 1.612 m; foresight at B: 2.034 m. What is the altitude of B? Is B higher or lower than A?

(b) If the leveler swaps backsight and foresight by mistake, what altitude does he get for B?

Your turn

One setup

(a) A has altitude 100.000 m. Backsight at A: 1.612 m; foresight at B: 2.034 m. What is the altitude of B? Is B higher or lower than A?

(b) If the leveler swaps backsight and foresight by mistake, what altitude does he get for B?

Answers

(a) $\Delta h = 1.612 - 2.034 = -0.422$ m; $H_B = 99.578$ **m**: B is *lower*.

(b) $\Delta h = +0.422$ m; $H_B = 100.422$ **m** — wrong by $2 \times 0.422 = 0.844$ m!

A sign error is a blunder: the adjustment (Class 3) finds it.

Small errors, long road

Random error

Each setup is off by a little, *sometimes up, sometimes down*, with no pattern. The errors partly compensate: the total grows **slowly**.



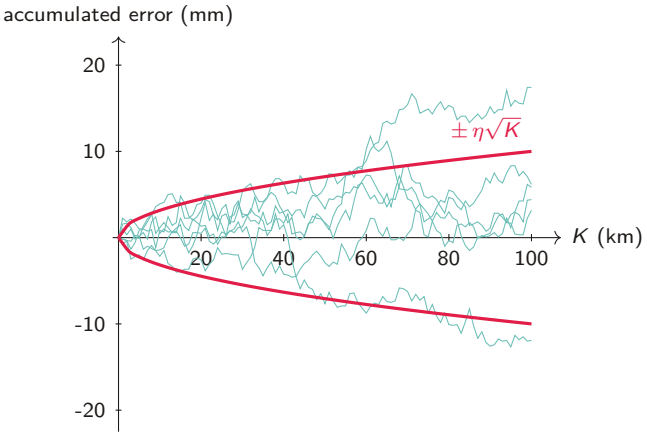
Systematic error

Each setup is off *always to the same side* (a rod with a "stretched" graduation, refraction, collimation). The errors *add up*: the total grows **fast**.



A 1000 km leveling line is thousands of setups. **Which one rules**? It depends on how many kilometers.

Random error: grows with the square root of the distance



Six simulated lines ($\eta = 1\,\text{mm}/\sqrt{\text{km}}$).

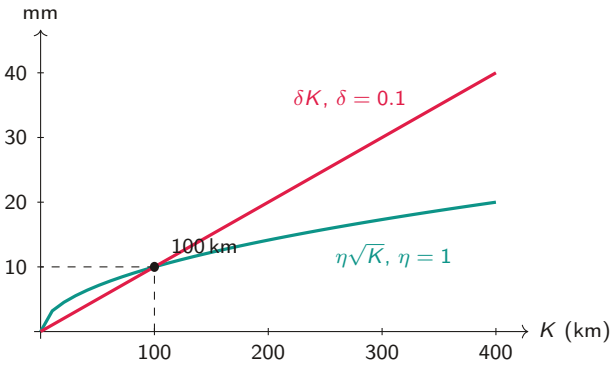
Each kilometer adds an error of *typical size* η, with a random sign. After K km, the typical deviation is

$$\sigma = \eta\sqrt{K}$$

η in $\text{mm}/\sqrt{\text{km}}$ is the "quality" of the leveling.

Quadrupling the distance **only doubles** the typical error.

Systematic error: grows with the distance



Square root ($\eta = 1\,\text{mm}/\sqrt{\text{km}}$) versus line ($\delta = 0.1\,\text{mm}/\text{km}$).

If each kilometer is off by δ *always to the same side*, the error over K km is

$$e = \delta\, K$$

Doubling the distance **doubles** the error.

The curves cross at $K^* = (\eta/\delta)^2 = 100$ **km**: from there on **the systematic wins**.

Side by side

K (km)	random $\eta\sqrt{K}$	systematic δK
4	2.0 mm	0.4 mm
100	10.0 mm	10.0 mm
400	20.0 mm	40.0 mm
2000	44.7 **mm**	200 **mm**

$\eta = 1\,\text{mm}/\sqrt{\text{km}}$; $\delta = 0.1\,\text{mm/km}$ (didactic).

Moral

On a short stretch, *random* error is what you see. On a 2000 km line, a *systematic* error of only 0.1 mm per kilometer makes 20 **cm**.

That is why the method — equal sights, invar, calibration, forward and back — is above all a **hunt for the systematic**.

Tolerance: when to redo a section

A section joins two benchmarks. Forward (1) and backward (2) height differences have opposite signs:

$$|d| = |\Delta h_1 + \Delta h_2| \leq 3\,\mathrm{mm}\sqrt{K}$$

K is the one-way length in km. Fundamental high-precision sections: at most 3 km.

K (km)	discrepancy limit
0.25	1.5 mm
1	3.0 mm
2	4.2 mm
3	5.2 mm

IBGE (1983), R.PR 22/1983, Table VI, items 3.1–3.3 and 4, p. 30.

Standard error after adjustment

The value $2\,\mathrm{mm}\sqrt{K}$ is the maximum standard error of an *adjusted line*. It evaluates a different quantity.

Lines and circuits

Forward/backward discrepancy of a **line**: $4\,\mathrm{mm}\sqrt{K}$. For a **circuit**, accumulated discrepancy divided by perimeter must not exceed 0.5 mm/km.

Your turn

Accept or redo?

(a) A section of $K = 2\,\text{km}$ gave forward $+8.912\,\text{m}$ and back $-8.904\,\text{m}$. What is the discrepancy? Does it pass the $3\sqrt{K}$ tolerance?

(b) On a 2000 km line, what is the typical random error ($\eta = 1$)? And the systematic one with $\delta = 0.1\,\text{mm/km}$? Which is larger, by what factor?

Your turn

Accept or redo?

(a) A section of $K = 2\,\text{km}$ gave forward $+8.912\,\text{m}$ and back $-8.904\,\text{m}$. What is the discrepancy? Does it pass the $3\sqrt{K}$ tolerance?

(b) On a 2000 km line, what is the typical random error ($\eta = 1$)? And the systematic one with $\delta = 0.1\,\text{mm/km}$? Which is larger, by what factor?

Answers

(a) $|8.912 - 8.904| = 8\,\textbf{mm}$;
$3\sqrt{2} = 4.2\,\text{mm}$. $8 > 4.2$: **redo** the section.

(b) $\sqrt{2000} = 44.7\,\text{mm}$;
$0.1 \times 2000 = 200\,\text{mm}$; the systematic one is 4.5 times larger.

Stop and think

A tempting idea

"Random error shrinks with averaging. So just *level the same line a hundred times* and the error vanishes."

What is right and what is wrong?

Stop and think

A tempting idea

"Random error shrinks with averaging. So just *level the same line a hundred times* and the error vanishes."

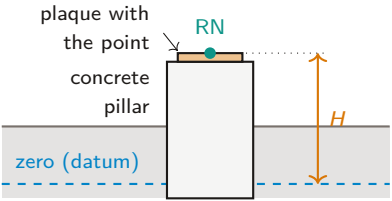
What is right and what is wrong?

Answer

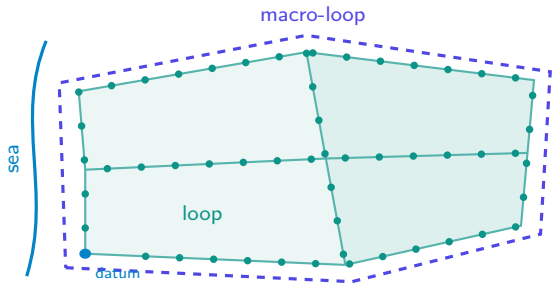
- The *random* part falls as $1/\sqrt{n}$: a hundred times gives **ten times** less.
- The *systematic* part **does not fall**: if the rod is "stretched", all hundred measurements are off equally.
- That is why everything is *varied*: forward and back, times, rods, instruments; and the rod is calibrated on a comparator.

The benchmark: the stone that keeps the altitude

- A **benchmark** (RN) is a mark — usually a metal plaque on a concrete pillar or on a structure — with a **known altitude**.
- They are placed *along roads and railways*, every few kilometers, so the leveler can arrive with a car and a rod.
- Each RN has a name (like **4X** or **9329T**) and a *sheet* in IBGE's **Geodetic Database** (BDG): description, position, altitude.



The network at three scales



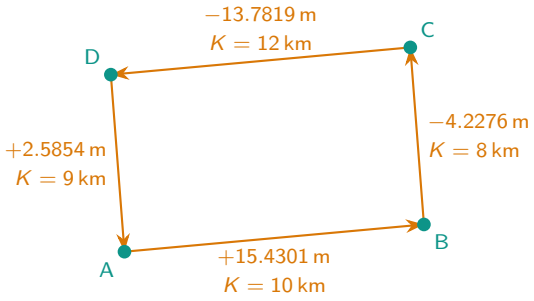
line benchmarks in sequence along a road, connected by sections leveled forward and back.

loop lines that *close a turn*: they return to the starting point.

macro-loop a big loop containing several: used in 1991 to split the network into adjustment pieces.

Sketch: the drawing does not represent any real region.

Loop misclosure



In this fictitious example, the corrected height differences should sum to **zero**.

Here: $w = \sum \Delta h = +6.0$ **mm**, perimeter $K = 39$ km.

With the corrections applied: limit $0.5K = 19.5$ mm. $|w|/K = 0.154$ mm/km: **passes the circuit check**.

IBGE (1983), VI.3.3, p. 30.

Necessary, not sufficient

Closing well does not guarantee there is no error: an error that depends on the height difference (like the "stretched" rod) *goes up and down* and cancels on the way back.

Network counts and their scope

Set	Benchmarks
AAGP (1992)	43 079
2011 adjustment	69 590
REALT-2018, principal	66 882

Principal block: 70 426 sections; 62 782 residuals in circuit-forming sections.

Which set is being counted?

66 882 counts the **principal block**, tied to Imbituba. The report treats lines excluded from that block and the Amapá network separately. This is not a national benchmark total.

The three sets have different scopes. A direct comparison does not measure network growth or benchmark loss.

IBGE (2011), pp. 2 and 12; IBGE (2019), pp. 35 and 40–44.

Your turn

Misclosure and turns

(a) A 100 km circuit has corrected misclosure $w = 21$ mm. Does it pass the 0.5 mm/km check? What is $|w|/K$?

(b) Compare two hypothetical lengths: 230 000 and 370 000 km. How many more turns around the Earth (40 075 km)?

Your turn

Misclosure and turns

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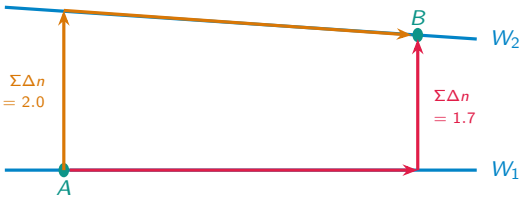
Answers

(a) $0.5 \times 100 = 50$ mm: **passes** ($21 < 50$);
$|w|/K = 21/100 = 0.21$ mm/km.

(b) $\dfrac{370\,000 - 230\,000}{40\,075} \approx 3.5$ turns.

Meeting the limit does not prove that systematic bias is absent.

Level surfaces are not parallel



Exaggerated. Along a level surface the height difference is zero.

The level measures the height difference between *neighboring* level surfaces. But the distance between W_1 and W_2 **changes from place to place**: where gravity is larger, it is smaller.

What is the same on both paths

$$\Delta W = g\,\Delta n$$

ΔW (the *geopotential* difference) is the same. $\Sigma\Delta n$ is not.

A perfect loop does not close at zero

Imagine a *perfect* leveler, with no error at all, who climbs a ridge by one path and comes down by another. The height difference the level "sees" in each stretch is $\Delta n = \Delta W/g$. If g is slightly different on the two paths, the loop **does not close**:

$$w \approx H\,\frac{\Delta g}{\gamma}$$

H is the climbed height and Δg the mean gravity difference between the paths.

H (m)	Δg (mGal)	w (mm)
500	20	10.2
1000	20	20.4
1000	50	51.1

Corrections and checks

The 20.4 mm effect precedes corrections. For 100 km, the corrected-circuit limit is 50 mm (0.5 mm/km).

IBGE (1983), pp. 28–30, VI.3.3.

($1\,\text{mGal} = 10^{-5}\,\text{m/s}^2$)

The geopotential number solves it

The way out (already seen in the geoid classes): **multiply each height difference by the gravity measured** on the stretch:

$$\Delta C = g\,\Delta n, \qquad C = \sum g\,\Delta n$$

- C **does not depend on the path**: every loop closes at zero (up to measurement error);
- unit: $\mathrm{m}^2/\mathrm{s}^2$ (about 9.8 times the height difference in meters);
- the altitude comes from C: $H^* = C/\bar{\gamma}$ (normal).

The price

To add up $g\,\Delta n$ you must know g *at each benchmark* — gravimetry on the leveling network.

And Brazil?

Until **2011**, without gravity measured at the benchmarks, only the correction with *theoretical gravity* was applied. Only **REALT-2018** brought *real* gravity: that is Class 3.

Stop and think

The riddle

"The leveler measured *perfectly*. Even so, the 100 km loop in rugged terrain closed at 20 mm. Who made the mistake?"

Stop and think

The riddle

"The leveler measured *perfectly*. Even so, the 100 km loop in rugged terrain closed at 20 mm. Who made the mistake?"

Nobody

- The error is in the *computation*, not in the measurement: $\sum \Delta n$ is not a physical quantity that closes around a loop.
- The quantity that closes is $\sum g\,\Delta n$ (geopotential number).
- On flat, short terrain the effect vanishes, which is why for decades it could be ignored. In a country of 8.5 million km^2, it cannot.

The four questions, answered

1. **How is the zero carried from the sea to a distant point?** By **geometric leveling**: setup by setup, with level and rod, $\Delta h = \text{back} - \text{fore}$, adding the height differences along a line of benchmarks.

2. **Where does the error come from and why does it grow with $\sqrt{K}$?** Each setup is a little off. If errors are random, the total grows as $\eta\sqrt{K}$; if systematic, as δK — and that one wins over long distances.

3. **What does the benchmark network look like?** Lines along roads, forming loops and macro-loops. The REALT-2018 principal block has 66 882 benchmarks; other sets are treated separately.

4. **Why does an error-free loop not close?** Because the level measures Δn and level surfaces are not parallel. What closes is the geopotential number, $\sum g\,\Delta n$.

Take home

Four ideas

1. The zero arrives by **leveling**: $\Delta h = \text{back} - \text{fore}$, setup by setup.
2. Random error: $\eta\sqrt{K}$. Systematic: δK. Section tolerance: $3\sqrt{K}$ mm (IBGE, 1983, p. 30).
3. The network is made of lines and loops; closing well is **necessary, not sufficient**.
4. $\sum \Delta n$ does not close; $\sum g\,\Delta n$ does.

Bridge to Class 3

Each section was leveled, but the network has *thousands* of loops that do not close exactly. Who decides the altitude of each benchmark? **The adjustment.**

And why has *the same benchmark* had several altitudes throughout history?

Class 3: The adjustments and REALT

Today's questions

1. Why has the *same* benchmark had several altitudes?
2. What does "adjusting" a leveling network mean — and how is it done?
3. How were Brazil's adjustments, from 1945 to 2011?
4. What did REALT-2018 change on the inside?

Deal

By the end of the class you will be able to answer all four — in your own words.

They come back on the last slide.

What the previous class left us

In Class 2, four ideas:

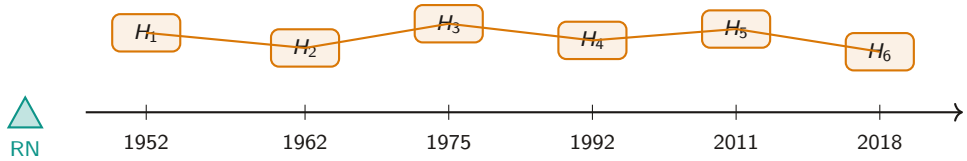
1. the zero arrives by **leveling**: $\Delta h = \text{back} - \text{fore}$;
2. each section has error: $\eta\sqrt{K}$ (random) and δK (systematic);
3. the network has **loops** that do not close exactly;
4. $\sum \Delta n$ does not close; $\sum g\,\Delta n$ does.

Today's question

Each loop closes with a *small disagreement*. In a country there are thousands of loops crossing each other. Which altitude do we give each benchmark? **One only**, which agrees with all measurements at once.

The same benchmark, several altitudes

the altitude of *one* benchmark, in six adjustments (illustrative heights)



New data

More lines, re-leveling, errors found and corrected.

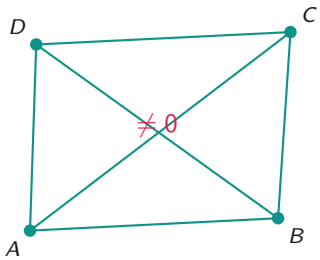
New method

Blocks → simultaneous adjustment; no gravity → with gravity.

New zero

Torres → Imbituba; Santana for Amapá.

Why adjust?



each loop closes with a disagreement

Six sections measured, **three** unknown altitudes (B, C, D; A is the datum). There are *more measurements than needed*: three extra.

Those *extra* measurements (redundancy) do not agree exactly with each other: each loop closes with a small disagreement.

Adjusting is...

finding **one single** altitude for each benchmark, spreading the disagreement fairly — and saying **how uncertain** each altitude is.

The model: each measurement is an equation

For the section from i to j, the measured height difference Δh_{ij} should be $H_j - H_i$. Since there is error, a **correction** v_{ij} is added:

$$\boxed{\Delta h_{ij} + v_{ij} = H_j - H_i}$$

Each section gets a **weight** $p_{ij} = \dfrac{1}{\eta^2 K_{ij}}$: long section = less reliable measurement = smaller weight.

Least squares

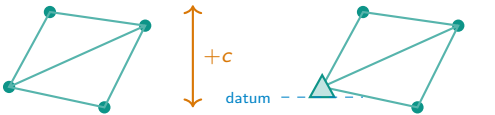
The altitudes are chosen that **minimize**

$$\sum p_{ij}\, v_{ij}^2$$

Large corrections are expensive (squared) and unreliable measurements are cheap (small weight).

It is the same method as in the `nivelamento/` simulator.

The datum enters as a fixed point



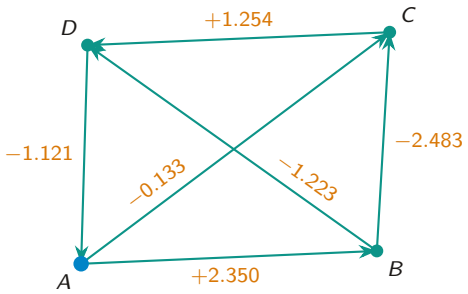
free network **constrained** network

All measurements are *differences*. Adding the same number c to all altitudes **changes no measurement**: the network "floats".

To fix it, **one point** is pinned to the datum: the Imbituba benchmark (4X) or the Santana one (9329T) is given an altitude. Then the solution is unique.

It is the same "rank defect" as the 3D datum in Part 1.

An example: the 4-node network



section	K (km)	σ (mm)
$A \to B$	4	2.00
$B \to C$	6	2.45
$C \to D$	5	2.24
$D \to A$	3	1.73
$A \to C$	5	2.24
$B \to D$	7	2.65

A is the datum: $H_A = 10.000\,\text{m}$. Height differences in meters; $\eta = 1\,\text{mm}/\sqrt{\text{km}}$ ($\sigma = \eta\sqrt{K}$).

The loop $A-B-D-A$ closes at $+6\,\textbf{mm}$; the other two close at zero.

The computations of the example

Build $N\,\hat{x} = b$ for the three unknowns (B, C, D), with $p = 1/K$:

$$N = \begin{pmatrix} 0.5595 & -0.1667 & -0.1429 \\ -0.1667 & 0.5667 & -0.2000 \\ -0.1429 & -0.2000 & 0.6762 \end{pmatrix} \mathrm{km}^{-1}$$

Each section ij puts $+p$ on the diagonal of i and of j and $-p$ off it: N is the "map" of the network.

Solving:

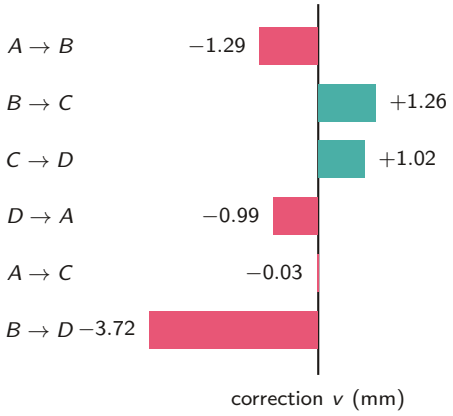
$$\hat{H}_B = 12.3487\,\mathrm{m}, \quad \hat{H}_C = 9.8670\,\mathrm{m}, \quad \hat{H}_D = 11.1220\,\mathrm{m}$$

No magic

It is only a system of 3 equations with 3 unknowns. In the real Brazilian network there are **tens of thousands**: the method is the same, the computer is what changes.

Full computation and check: `sgb_altimetria_numeros.py` and appendix A2.

What the adjustment did



The 6 mm disagreement in $A-B-D-A$ was **spread**: the largest correction fell on $B \to D$, the longest section ($K = 7$ km, smallest weight).

The **variance factor** $\hat{\sigma}_0 = 1.03$ is almost 1: the data agree with $\eta = 1$. The altitudes come out with $\sigma \approx 1.4$ to 1.6 mm.

Check: the example was *made up* from known altitudes (12.348; 9.868; 11.121 m). The adjustment came within about **1 mm** of each.

Your turn

Adjusting one loop

A loop of three sections, with $K = 2$, 3 and 5 km, closed at $w = +5$ mm. The weights are $p = 1/K$. How much is each section corrected so that the loop closes at zero?

Hint: the correction is proportional to K.

Your turn

Adjusting one loop

A loop of three sections, with $K = 2$, 3 and 5 km, closed at $w = +5$ mm. The weights are $p = 1/K$. How much is each section corrected so that the loop closes at zero?

Hint: the correction is proportional to K.

Answer

$v_i = -w \dfrac{K_i}{\sum K}$ with $\sum K = 10$:

$v_1 = -1.0$, $v_2 = -1.5$,
$v_3 = -2.5$ mm.

Sum: -5 mm $= -w$: the loop closes. Whoever has more kilometers **absorbs more**.

Stop and think

A tempting idea

"After the adjustment, the loops close at zero and the altitudes have an uncertainty of a few millimeters. So the *errors are gone*."

What is wrong?

Stop and think

A tempting idea

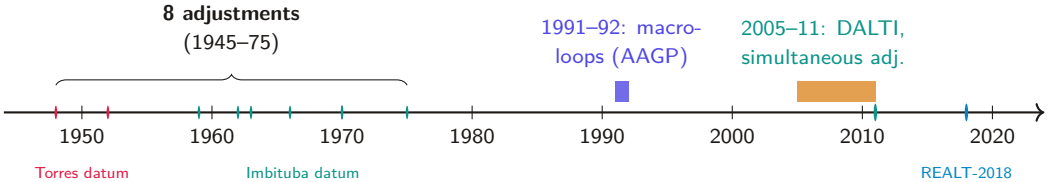
"After the adjustment, the loops close at zero and the altitudes have an uncertainty of a few millimeters. So the *errors are gone*."

What is wrong?

Answer

- The adjustment **spreads** the disagreement; it does not make it disappear.
- It reduces the *random* error; the *systematic* one that closes well in a loop is still there.
- A *blunder* can be spread over the neighbors and hidden — which is why the residual test (*data snooping*) is run, as in the simulator.

The history of Brazil's adjustments



Up to 1975

Eight adjustments; from the fourth onward, by blocks.

1991–2011

From macro-loops to a *simultaneous* adjustment.

2018

The same kind of adjustment, now with *real gravity*.

1945–1975: eight adjustments

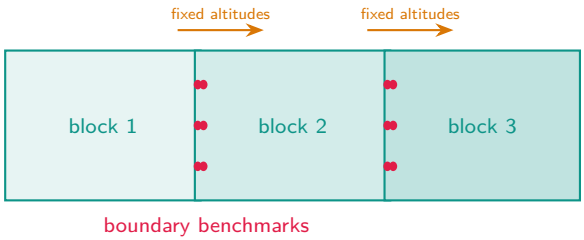
- Between 1945 and 1975, IBGE made **eight adjustments**: 1948, 1952, 1959, 1962, 1963, 1966, 1970 and 1975.
- Only the **first three** used *all* the available observations.
- From the fourth on, with too many observations, the network was **split into blocks**: some benchmarks kept a *fixed* altitude, computed in the previous blocks.
- Those of 1948 and 1952 were on the **Torres** datum; the six from 1959 to 1975, on **Imbituba**.

Why blocks?

The computation used the means of the day: the network grew faster than the system of equations could be solved at once. Splitting into blocks was the practical way out.

Source: IBGE (2011), pp. 10–12; Siqueira et al. (2018), slides 3–4.

Blocks in cascade



1. Block 1 is adjusted.
2. The boundary benchmarks become *fixed points* in block 2.
3. And so on.

Block 1 **never** benefits from the measurements of the following blocks, and the leftover disagreement gets "pushed" to the end of the cascade.

A small test: cascade × simultaneous adjustment

With the 4-node network, we do *two* adjustments:

- **simultaneous**: all 6 sections at once;
- **cascade**: block 1 (A, B, C: sections AB, BC, AC); then B and C *fixed*, and block 2 (sections CD, DA, BD) finds D.

	H_B (m)	H_C (m)	H_D (m)
simultaneous	12.3487	9.8670	11.1220
cascade	12.3500	9.8670	11.1223
difference (mm)	+1.3	+0.0	+0.3

Two differences

- The *altitudes* differ: B changed 1.3 mm, with no new measurement — only because of the order of computation.
- The *uncertainty* of D comes out **optimistic**: $\sigma_D = 1.22$ mm in the cascade (which treats B and C as error-free) versus 1.38 mm in the simultaneous one — 12 % smaller.

1991–92: macro-loops

- The work begun in **1988** divides the network into **macro-loop** sub-networks to reduce distortions.
- The result is the **AAGP**, the Preliminary Global Altimetric Adjustment: still *not simultaneous*.
- The BDG uses **1 December 1992** as the AAGP calculation reference date: 43 079 benchmarks. Isolated adjustments and spurs followed.

IBGE (2011), pp. 11–12.

The macro-loop, again

A macro-loop is a *large* loop containing several. Adjusting each sub-network before joining them removes the worst pushes of the cascade, but still leaves a **seam** between the parts.

2005–2011: the simultaneous adjustment

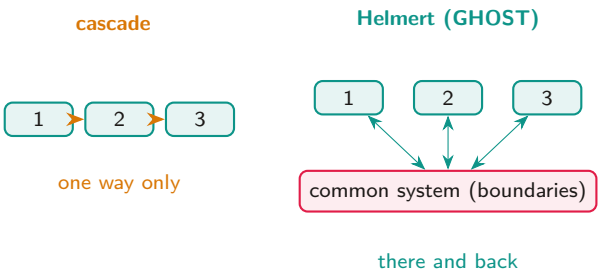
- IBGE's altimetry team (**DALTI**, in the Geodesy Coordination) starts in **2005** an adjustment of **everything at the same time**.
- Result, released in **June 2011**: altitudes for 69 590 **benchmarks**, from more than 74 000 **observations** made since 1945.
- Software: **GHOST** (Canadian), made to adjust huge networks.
- Still *without measured gravity*: only the correction with theoretical gravity.

66 years later

From 1945 to 2011 is 66 **years**. The new processing returns to simultaneous adjustment after the partitioned phase; the first three historical adjustments were global too.

Source: IBGE (2011), presentation, p. 2, and introduction, p. 7.

Blocks that talk



In **Helmert blocking**, the blocks are solved *inside* and only the *boundary* unknowns form a common system, solved at once; then each block gets the answer back.

The point

The solution is **exactly** that of the simultaneous adjustment: the block is only a trick of *computation*. In the cascade, it is an *approximation*.

Your turn

How many loops does Brazil have?

The 2011 adjustment had 69 590 benchmarks and more than 74 000 observations (sections). There are **two** network blocks (Imbituba and Santana), each with one fixed point. How many unknowns? How many observations *extra*, at least (the redundancy)? What fraction is that?

Your turn

How many loops does Brazil have?

The 2011 adjustment had 69 590 benchmarks and more than 74 000 observations (sections). There are **two** network blocks (Imbituba and Santana), each with one fixed point. How many unknowns? How many observations *extra*, at least (the redundancy)? What fraction is that?

Answers

$69\,590 - 2 = 69\,588$ unknowns.

$r \geq 74\,000 - 69\,588 = 4412$.

$4412/74\,000 \approx 6\,\%$: only *one in every 17* sections "closes" a loop.

The network is almost a *tree*: each loop is worth a lot.

REALT-2018: what changed

1. Real gravity

Observed gravity, supplemented by **interpolation**, enters the calculation at all benchmarks. In 2011 the correction used *theoretical* gravity.

2. A different unknown

The adjustment computes **geopotential numbers** C (m^2/s^2), not altitudes. The *normal* height comes afterward: $H^* = C/\bar{\gamma}$.

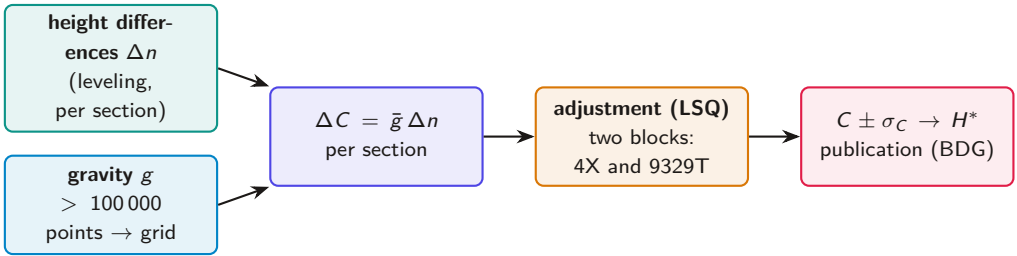
3. Revised data

Complete review of the databases, new leveling and gravimetry observations, and inconsistencies corrected.

What did not change: the two zeros. REALT-2018 kept the Imbituba and Santana datums; what changed is what is *propagated* from them.

Released on 30/07/2018; 2nd edition of the report on 30/09/2019.

The data path in REALT-2018



Up to 2011: $\Delta n \rightarrow$ correction with theoretical gravity $\rightarrow$ adjustment of altitudes. **In 2018**: Δn and observed/interpolated $g \rightarrow \Delta C \rightarrow$ adjustment of $C \rightarrow H^*$.

Where gravity comes in

For each section ij, the geopotential difference is the height difference *times the mean gravity* between its two ends:

$$\Delta C_{ij} = \bar{g}_{ij}\,\Delta n_{ij}, \qquad \bar{g}_{ij} \approx \frac{g_i + g_j}{2}$$

Both **observed and interpolated** gravity values were used. Interpolation combined **more than 100 000 points** from IBGE and BNDG using Geogrid_7c.

IBGE (2019), pp. 29–32; Siqueira et al. (2018), slide 20.

The same *gravimetry* that feeds the geoid now also feeds the leveling network.

How much does gravity vary?

Over the whole Earth, g ranges from 9.764 to 9.834 m/s², about 0.7 % (values quoted by IBGE). It looks small, but 0.7 % of an altitude of 1000 m is **7 m**: that is why gravity can *not* be treated as constant.

What matters for leveling is the variation *between paths* (Class 2); the variation with latitude is already handled by normal gravity.

From C to the normal height

The result of the adjustment is the geopotential number C of each benchmark, with its uncertainty σ_C. The altitude comes out by **one division**:

$$H^* = \frac{C}{\bar{\gamma}}$$

$\bar{\gamma}$ is the mean *normal* gravity along the benchmark's vertical (ellipsoid formula, nothing measured).

Example (fictitious benchmark)

$C = 68.392\,\mathrm{m^2/s^2}$

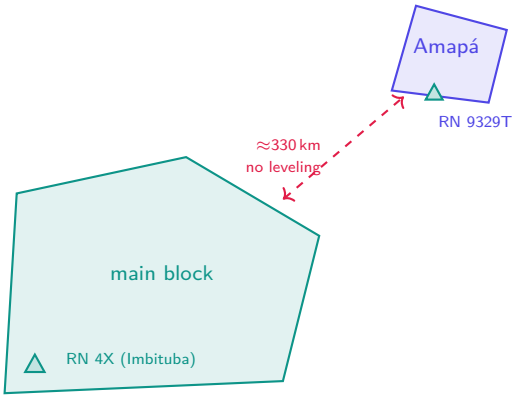
$\bar{\gamma} = 9.78\,\mathrm{m/s^2}$

$$H^* = \frac{68.392}{9.78} = 6.993\,\mathbf{m}$$

Rule of thumb: $\mathbf{1\,m} \approx 9.8\,\mathbf{m^2/s^2}$.

Example values, made up; not from a real benchmark.

Two blocks, two datums, one method

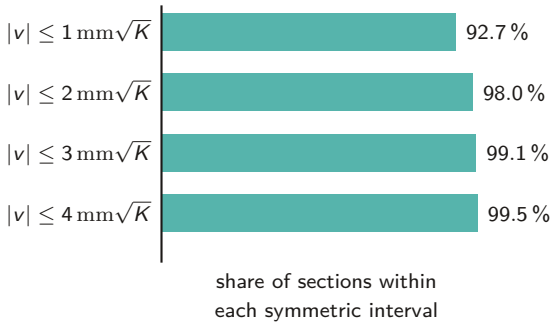


REALT-2018 makes **two adjustments**:

- the **main block**, pinned to benchmark **4X** (Imbituba);
- the **Amapá** network, pinned to benchmark **9329T** (Santana).

There is no leveling between them. Each block has *its* zero — and the altitudes of one are not directly comparable to those of the other.

REALT-2018: principal-block residuals



IBGE presents height-difference corrections as a function of length: $v/\sqrt{K}$, with K in km. These percentages describe the **principal block of REALT-2018**.

Adjustment results

These frequencies assess observation consistency in the adjusted model. Field acceptance uses section, line and circuit tolerances. Height uncertainty also requires the covariance information.

IBGE (2019), REALT-2018, 2nd ed., p. 36, Figure 16; Siqueira et al. (2018), slide 31.

What changed at the benchmarks, in practice

Isolating gravity integration

With the same data and geometry, integrating real gravity produced differences between -15 and $+5$ mm at **94 %** of benchmarks.

The whole readjustment

About 50 000 benchmarks (76 %) changed by $+20$ to $+30$ cm, in much of the North, Northeast, Center-West and Southeast.

In IBGE's controlled test, the isolated gravity effect was millimetric at 94 % of benchmarks. Data cleaning and network geometry explain much of the difference in the complete readjustment.

Source: IBGE (2019), p. 37, Figure 18; p. 40, Figure 20. These are two different comparisons.

Your turn

From C to H^*

(a) A benchmark has $C = 68.392\,\mathrm{m}^2/\mathrm{s}^2$ and $\bar{\gamma} = 9.78\,\mathrm{m}/\mathrm{s}^2$. What is the normal height?

(b) If $\bar{\gamma} = 9.80$ were used by mistake, how much would it change?

(c) And for a benchmark with $H^* = 1500\,\mathrm{m}$, with the same relative error?

Your turn

From C to H^*

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(b) If $\bar{\gamma} = 9.80$ were used by mistake, how much would it change?

(c) And for a benchmark with $H^* = 1500\,\mathrm{m}$, with the same relative error?

Answers

(a) $68.392/9.78 = 6.993\,\mathbf{m}$.

(b) $68.392/9.80 = 6.979\,\mathrm{m}$: changes by $14\,\mathbf{mm}$.

(c) The error is *relative*: $0.2\,\% \times 1500 \approx 3\,\mathbf{m}$. That is why $\bar{\gamma}$ is computed carefully, point by point.

Stop and think

One project, two versions

A drainage project uses benchmark P with the altitude from the *2011 adjustment* and benchmark Q with the altitude from *REALT-2018*. Both are official. What can go wrong?

Stop and think

One project, two versions

A drainage project uses benchmark P with the altitude from the *2011 adjustment* and benchmark Q with the altitude from *REALT-2018*. Both are official. What can go wrong?

Answer

- If the region rose about $+25$ cm from one version to the other, mixing the two creates a **false step** of decimeters between P and Q — and the water flows down the step.
- The *difference* between benchmarks of the *same* version is what is useful.
- Rule: always note **which adjustment** (year and datum) gave the altitude.

The four questions, answered

1. **Why has the same benchmark had several altitudes?** Because each adjustment uses **new data**, a **new method** and sometimes a **new zero** — and the altitude is the *result* of the adjustment, not a property of the stone.
2. **What is adjusting a network?** Finding one altitude for each benchmark by **least squares**: $\Delta h + v = H_j - H_i$, weight $1/K$, one fixed point at the datum; spreading the disagreement and giving the uncertainty.
3. **How were the adjustments from 1945 to 2011?** Eight adjustments, the first three global (1948–75), macro-loops (1991–92) and the return to **simultaneous adjustment** (2005–11): 69 590 benchmarks, more than 74 000 observations.
4. **What did REALT-2018 change?** The unknown became the **geopotential number**, with **real gravity** at all benchmarks, revised data and two blocks (Imbituba and Santana); $H^* = C/\bar{\gamma}$ is derived.

Take home

Four ideas

1. Altitude **is the result of an adjustment**, not a property of the stone.
2. Adjusting $=$ least squares with weights $1/K$ and a fixed datum.
3. Cascade of blocks $\neq$ simultaneous adjustment (Helmert blocking is *exact*).
4. REALT-2018: C with real gravity; $H^* = C/\bar{\gamma}$; two zeros.

Bridge to Class 4

The two zeros remained. But *how do they relate*? Today's tide gauges measure the sea with GNSS next to them. And the future is a **global** zero: the IHRS.

Class 4: The zero, now and next

Today's questions

1. How does IBGE watch today's sea against yesterday's zero?
2. Do Brazil's two zeros agree? How much do they differ?
3. What do SIRGAS and the IHRS propose for the future?
4. In practice: level, use GNSS and a model — and what to do if the benchmark disappears?

Deal

By the end of the class you will be able to answer all four — in your own words.

They come back on the last slide.

What the previous class left us

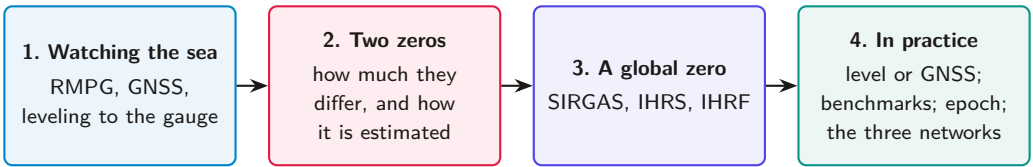
In Class 3, four ideas:

1. the altitude is the **result of an adjustment**;
2. adjusting is least squares, with weights $1/K$ and one point fixed at the datum;
3. the history was: blocks $\rightarrow$ macro-loops $\rightarrow$ simultaneous adjustment (2011);
4. REALT-2018 adjusts C with real gravity; $H^* = C/\bar{\gamma}$.

What stayed open

The two **zeros** were kept — Imbituba and Santana — and the sea of 1949–57 is no longer today's. How do we know *how much* they are off? And how do they relate to each other?

The thread of the class



Why watch the sea?

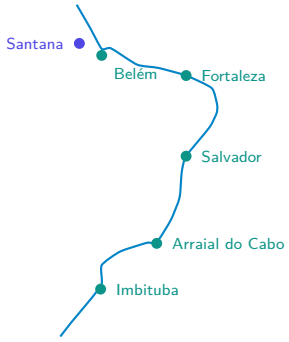
- The mean sea levels that define the benchmarks were computed with data from **different epochs**, often decades ago.
- That is why IBGE created the **RMPG**, the Permanent Tide Gauge Network for Geodesy: an idea from **1996** (UFRJ and UFPR), run by the Geodesy Coordination since **2001**.
- Goal: to relate the vertical datum (Imbituba, Santana) to *today's sea level* and to other coastal reference levels.

A series of more than one cycle

The consolidated RMPG series runs from **July 2001 to December 2024**: about 23.5 **years**, more than one 18.6-year nodal cycle.

Public data, updated daily; weather (meteorological) data have been published since 1/4/2022.

The six stations

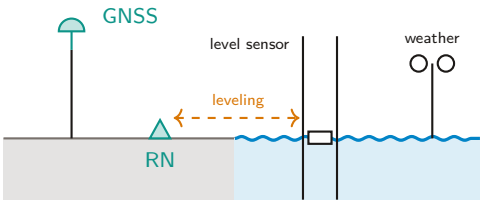


IBGE network in Technical Note 01/2024:

Imbituba (SC)	the **datum** of the South and Center of the country
Arraial do Cabo (RJ)	Southeast coast
Salvador (BA)	Northeast coast
Fortaleza (CE)	Northeast coast
Belém (PA)	Amazon mouth
Santana (AP)	the **datum** of Amapá

Partners: Navy (DHN), UFPR, Univ. of Hawaii, Petrobras, FURG and port authorities. An older source mentions five stations.

What is in a station



level the sensor measures the sea *relative to the ground*.

leveling ties the sensor's zero to an SGB benchmark: *so the reading becomes altitude*.

GNSS the vertical motion of the *ground*.

weather wind and pressure help understand jumps in the level.

IBGE's manual (Technical Manual No. 11, 2010) also covers gravimetry at the stations.

From the sensor reading to the sea's altitude

An *invented* example, with the station's benchmark:

altitude of the benchmark (SGB)	3.250 m
the benchmark is above the sensor's zero	4.100 m
$\Rightarrow$ altitude of the sensor's zero	-0.850 m
mean sea level reading	$+0.995$ m
local MSL, in SGB altitude	**$+0.145$ m**

What it means

In this example, the sea *of that window* is 14.5 cm *above* the datum's zero. That is exactly what the RMPG computes, station by station, and what each port's "reference levels" documents contain.

Each port is a case

The value changes from station to station: sea surface topography, epoch and ground motion.

GNSS at the tide gauge: the ground rules

The GNSS antenna gives the **vertical velocity of the ground** (mm/year), relative to the Earth's center. Small per year, *large over decades*:

velocity	in 10 years	in 30 years
$+3$ mm/yr	$+30$ mm	$+90$ mm
-3 mm/yr	-30 mm	-90 mm

Separating sea and ground

The tide gauge sees *sea minus ground*. GNSS sees only the ground. The difference gives the **real sea**.

Imbituba

A UFPR study (2017) found, with GNSS campaigns, a sinking of the tide gauge of about 3 mm/year — over 72.5 years that would be ≈ 22 cm. Value from one study; short campaigns.

Your turn

The sea at a station

At a station, the benchmark has altitude 2.870 m and is 3.540 m above the sensor's zero. The mean sea reading, over a 19-year window, was $+0.812$ m above the sensor's zero. What is the altitude of the local MSL in the SGB system? Is it above or below the datum?

Your turn

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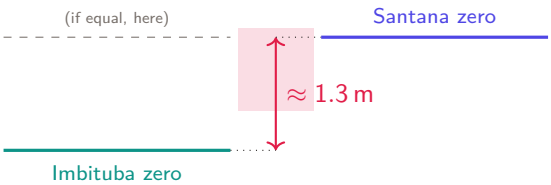
Answer

Sensor's zero:
$2.870 - 3.540 = -0.670$ m.

MSL: $-0.670 + 0.812 = +0.142$ **m**.

It is 14.2 **cm above** the datum's zero — consistent with a sea that has risen since the datum's epoch.

Do the two zeros agree?



There is no leveling between the two networks. So the difference between the zeros can only be *estimated*, with GNSS, gravimetry and models.

A study with a global model from the GOCE satellite found the Santana network 1.30 **m** above the Imbituba one; other findings range from 0.45 to 1.40 m, depending on the strategy.

Values from studies (2013–2019), not official; from the search summary (appendix A8).

How to estimate the difference without leveling

At each benchmark with GNSS, the satellite height is compared with the network altitude and with a quasigeoid *model*:

$$r_i = h_i - H_i^* - \zeta_i^{\text{model}}$$

If the network's zero coincided with the model's, the r_i would scatter around *zero*. The **mean** of the r_i in each network is the *offset of its datum* δ:

$$\delta_S - \delta_I = \overline{r}_S - \overline{r}_I$$

The idea

It is the same *GNSS/leveling* computation of the quasigeoid class, used for a new purpose: measuring the **lag between zeros**, instead of testing the model.

The limit

The result is only as good as the *model*: where gravimetry is sparse (the Amazon), the model error can be decimeters.

Why do the studies differ?

- Each study uses a different **global model** and **strategy** (GNSS and model; leveling with gravimetry; physical connection).
- In the region between the two networks (the Amazon mouth and Marajó), gravimetry is **scarce** and leveling cannot pass.
- The two datums are MSL of different **epochs and windows** (9 years $\times$ 2 years).

A yardstick of the problem

For the South American vertical systems as a whole, global estimates give an accuracy of **centimeters** where data are good, **decimeters** where they are sparse and up to **1 m** in extreme cases.

Values from a DGFI-TUM presentation (search summary).

Your turn

The lag between networks

At three benchmarks of the Imbituba block, $r = h - H^* - \zeta$ gave 0.10, 0.14 and 0.12 m. At three benchmarks in Amapá: 1.38, 1.44 and 1.42 m (made-up numbers). What is the lag $\delta_S - \delta_I$?

Your turn

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Answer

$\overline{r}_I = 0.120\,\text{m}$; $\overline{r}_S = 1.413\,\text{m}$.

$\delta_S - \delta_I = 1.293\,\mathbf{m} \approx 1.3\,\text{m}$.

Within each network the r agree with each other (a few cm); between networks there is a *step* of ≈ 1.3 m.

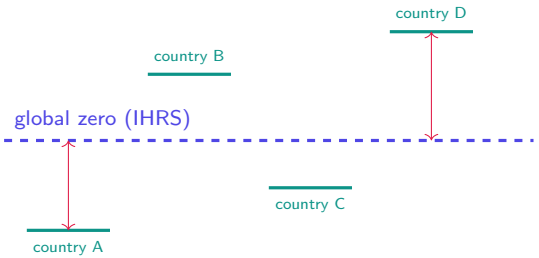
What to do meanwhile?

1. **Do not mix** altitudes of the two networks without saying so: the difference may be *over a meter*.
2. **hgeoHNOR2020** delivers H^* in the *local* datum: it was made to agree with the official benchmarks, each in its own zero.
3. A project crossing the Amazon must **state** the datum of each stretch.

The classic mistake

Adding altitudes of Macapá (Santana datum) to elevations in Pará (Imbituba datum) and thinking "the job closes". It closes on paper; on the ground, water flows down a hidden step.

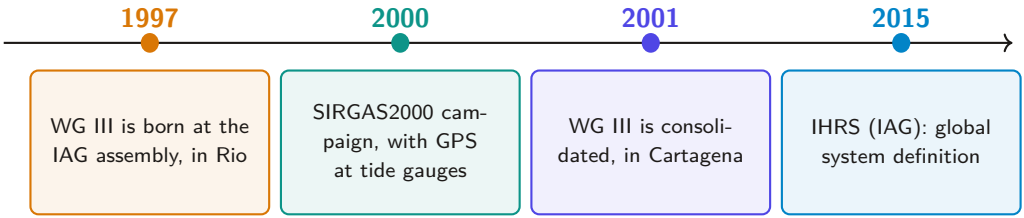
Why not simply change the zero?



Each country has *its* local zero; between neighbors the difference can reach 1–2 m.

- Changing the zero changes **all** altitudes: maps, projects, laws.
- A *better* zero was missing: only in **2015** did the IAG define the **IHRS**, with a *geopotential value* $W_0 = 62\,636\,853.4\,\mathrm{m}^2/\mathrm{s}^2$, not a tide gauge.
- The route is to **link** each datum to the IHRS (one offset per datum), and only then decide what to change.

SIRGAS and WG III



SIRGAS's **Working Group III** (Vertical Datum) was born in **1997** to *unify* the heights of the Americas: each country had its own tide gauge, epoch and window.

What the IHRF asks of each country

The vertical realization of SIRGAS will be a **regional densification of the IHRF**, with:

- the SIRGAS GNSS reference stations;
- the **main tide gauges** of each country;
- **border** stations, linking the first-order leveling networks of neighbors.

The physical component uses **geopotential numbers** referred to a common W_0, at a specified epoch. Leveling and gravimetry help connect the local datums.

SIRGAS, Working Group III, vertical definition and realization.

And Brazil?

REALT-2018 was adjusted in *geopotential numbers* precisely to prepare the network for the IHRS. Imbituba and Santana are main tide gauges. The connection requires expressing both datums relative to the global W_0.

In meters and in $\mathrm{m}^2/\mathrm{s}^2$

$1.30\,\mathrm{m} \times 9.78 \approx 12.7\,\mathrm{m}^2/\mathrm{s}^2$.

Your turn

Meters and geopotential

(a) How many m^2/s^2 is 0.15 m worth (the MSL height above the datum, in the earlier example)? Use 9.78.

(b) A datum has an offset of $12.7\,m^2/s^2$ from W_0. How many meters?

Your turn

Meters and geopotential

(a) How many m^2/s^2 is 0.15 m worth (the MSL height above the datum, in the earlier example)? Use 9.78.

(b) A datum has an offset of 12.7 m^2/s^2 from W_0. How many meters?

Answers

(a) $0.15 \times 9.78 = 1.47\,m^2/s^2$.

(b) $12.7/9.78 = 1.30\,\mathbf{m}$.

A step of *a meter and a half* is only 13 to 15 geopotential units: that is why IHRS uncertainties come in m^2/s^2.

Level, or GNSS and a model?

Task	What matters	Tool
drainage, canal, sewer	**small differences** between nearby points	leveling from nearby benchmarks
map; long project; point far from benchmarks	**absolute** altitude in the SGB	GNSS + hgeoHNOR2020 (quasigeoid class)
coastal work, port, flood	relation to the **local sea**	RMPG and IBGE's reference-levels documents

Rule of thumb

Leveling is good for *differences* between neighboring points; GNSS with a model is good for *reaching* an absolute value where there is no benchmark. In serious works, both are used: GNSS to get to the area, leveling for the work.

The life of benchmarks

A benchmark is a stone by the road. Roadworks, expansions and vandalism **destroy** them; the ground may **sink**. Before using one:

1. read its **sheet** in the BDG (description, epoch, altitude);
2. confirm *in the field* that the benchmark exists and matches the description;
3. tie to **two** benchmarks and compare the measured height difference with the published one.

A field test (made-up numbers)

Published: $H_1 = 12.345$ and $H_2 = 20.100\,\text{m}$; $\Delta H = 7.755\,\text{m}$. New *corrected* difference: 7.759 m; discrepancy 4 mm. With illustrative combined uncertainty $\sigma_r = 5\,\text{mm}$, $|r|/\sigma_r = 0.8$. σ_r combines the new measurement with benchmark uncertainties and covariance. The section forward/backward limit evaluates a different quantity.

Three dates in every altitude

The datum's epoch

the MSL of 1949–57 (Imbituba) or 1957–58 (Santana)

The network version

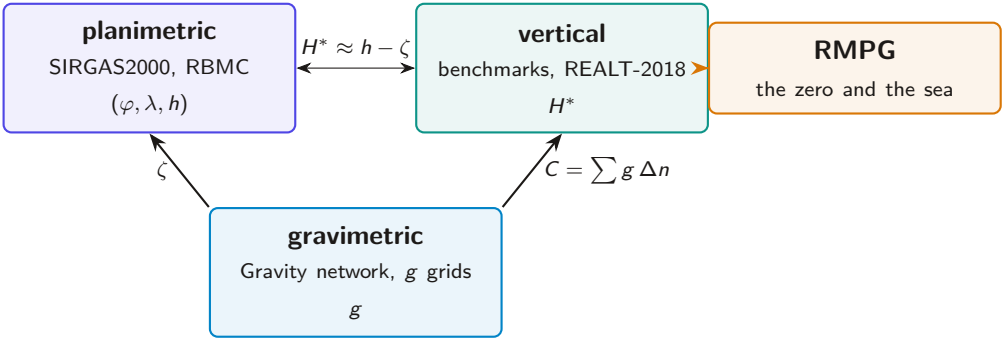
the adjustment: 2011, REALT-2018, 2nd edition 2019

The day of the measurement

the ground has moved since: mm per year

An honest altitude states all three: "$H^* = 6.993$ **m, Imbituba datum, REALT-2018, measured in 2026**". Without that, it is only as good as the three look alike.

The three networks of the SGB



The three networks help each other: **gravity** gives the geoid and the geopotential number; **GNSS** gives h; the **leveling network** gives H^*; and the **RMPG** ties the zero to the sea.

Stop and think

The closing question

"IBGE has a datum at Imbituba, another at Santana, a new network of 2018 and an IHRS on the horizon. For a *house* in Florianópolis, none of this matters, right?"

What would you answer?

Stop and think

The closing question

"IBGE has a datum at Imbituba, another at Santana, a new network of 2018 and an IHRS on the horizon. For a *house* in Florianópolis, none of this matters, right?"

What would you answer?

Answer

- For the house *alone*, almost nothing: the work uses the difference with a nearby benchmark.
- But **the sewer, the street and the river** use altitudes of the same system: if each project chose a zero, nothing would fit.
- The work of datums, adjustments and REALT is so that **everyone has the same zero** — and so that it is known *which* one it is.

The four questions, answered

1. **How does IBGE watch today's sea?** With the **RMPG**: six tide gauges with a level sensor, weather, **GNSS** and leveling to a benchmark. This gives the local MSL *in SGB altitude*, station by station, and the ground motion.
2. **Do the two zeros agree?** It is not known for sure: **studies** give 0.45 to 1.40 m (about 1.3 m, Santana above). With no leveling between the networks, it can only be *estimated*.
3. **What do SIRGAS and IHRS propose?** A **global** zero, a geopotential value (W_0), and a frame of stations (**IHRF**) with geopotential numbers and main tide gauges; WG III has worked on it since 1997.
4. **And in practice?** Level for *differences*, GNSS with a model to *get there*; check the benchmark in the field; and always state **datum, version and date**.

Summary of Part 2

Class 1 · The zero

Altitude is the reading of a ruler. The zero is **one MSL, at one station, in one window**: Torres, then Imbituba (1949–57) and Santana (1957–58).

Class 2 · Leg by leg

The zero arrives by **leveling**: $\Delta h = \text{back} - \text{fore}$, error $\eta\sqrt{K}$ (random) or δK (systematic), network of benchmarks, $\sum g\,\Delta n$ closes.

Class 3 · The adjustments

The altitude is the **result of an adjustment**: blocks $\rightarrow$ macro-loops $\rightarrow$ simultaneous (2011) $\rightarrow$ REALT-2018 with C and real gravity.

Class 4 · The zero, now and next

The RMPG watches the sea; the **two zeros** differ by about 1.3 m; the future is the **IHRS**; and every altitude has **three dates**.

Part 1: **where** (SIRGAS2000). Part 2: **how high** (the zero, leveling, REALT).

Take home

Four ideas

1. The RMPG ties **today's sea** to yesterday's zero.
2. The two zeros differ by about $1.3\,\mathbf{m}$ (an estimate, not official).
3. The future is a **global** zero: the IHRS, with $W_0 = 62\,636\,853.4\,\mathrm{m}^2/\mathrm{s}^2$.
4. Every altitude states **datum, version and date**.

Where to read more

The computations of each slide are in `sgb_altimetria_numeros.py`; the formulas and the history, in the appendix. The geoid and the quasigeoid have their own classes: `geoide_e_alturas`, `quase_geoide` and `gravimetria`.

Technical Appendix

Formulas, computations, timeline, glossary and sources

A1 · Geopotential number and kinds of height

Formulas only; the derivation is in the geoid classes (`geoide_e_alturas`, `quase_geoide`).

Geopotential number

$$C_P = W_0 - W_P = \int_0^P g \, \mathrm{d}n \approx \sum_i \bar{g}_i \, \Delta n_i$$

Unit: $\mathrm{m}^2/\mathrm{s}^2$ (1 m of height $\approx$ 9.8 $\mathrm{m}^2/\mathrm{s}^2$).
Path-independent; closes around any loop.

Three heights from C

$$H^* = \frac{C}{\bar{\gamma}} \quad \text{(normal)}$$

$$H = \frac{C}{\bar{g}} \quad \text{(orthometric)}$$

$$H^{\mathrm{dyn}} = \frac{C}{\gamma_{45}} \quad \text{(dynamic)}$$

$\bar{\gamma}$: mean *normal* gravity along the normal plumb line; $\bar{g}$: mean *actual* gravity along the plumb line; γ_{45}: normal gravity at 45° latitude (fixed value).

In Brazil, since 30/07/2018, the official heights are H^* (REALT-2018).

A2 · Least squares for leveling

Model and solution

For each section ij: $\Delta h_{ij} + v_{ij} = H_j - H_i$, or $\ell + v = A\,x$, with A having rows $(\dots, -1, \dots, +1, \dots)$ and $P = \mathrm{diag}(1/K_{ij})$ (η^2 factors out).

$$N = A^{\mathsf{T}} P A, \quad b = A^{\mathsf{T}} P \ell, \quad \hat{x} = N^{-1} b$$

N is the *weighted Laplacian* of the network graph (without the rows and columns of fixed points).

Precision

$$\hat{\sigma}_0^2 = \frac{v^{\mathsf{T}} P v}{m - n}, \qquad \Sigma_{\hat{x}} = \hat{\sigma}_0^2\, \eta^2 N^{-1}$$

m observations, n unknowns, $r = m - n$ degrees of freedom. Global test: $r\,\hat{\sigma}_0^2 \sim \chi^2_r$ if the stochastic model holds (the simulator's "Global Test").

Standardized residuals (*data snooping*): $w_i = v_i / \sigma_{v_i}$, with $Q_{vv} = P^{-1} - A N^{-1} A^{\mathsf{T}}$.

Free network: with no fixed point, N is singular, with null space $(1, \dots, 1)^{\mathsf{T}}$ (adding c to all altitudes): rank defect 1 per connected component. **Constraint**: fixing H_A removes the defect.

A2 · The 4-node example, in full

section	Δh (m)	K (km)	v (mm)
$A \to B$	$+2.350$	4	-1.29
$B \to C$	-2.483	6	$+1.26$
$C \to D$	$+1.254$	5	$+1.02$
$D \to A$	-1.121	3	-0.99
$A \to C$	-0.133	5	-0.03
$B \to D$	-1.223	7	-3.72

$H_A = 10.000\,\text{m}$ fixed; $p = 1/K$; $m = 6$, $n = 3$, $r = 3$.

$$N = \begin{pmatrix} 0.5595 & -0.1667 & -0.1429 \\ -0.1667 & 0.5667 & -0.2000 \\ -0.1429 & -0.2000 & 0.6762 \end{pmatrix} \text{km}^{-1}$$

$$b = \begin{pmatrix} 3.6761 & 1.3088 & 3.7831 \end{pmatrix}^{\mathsf{T}} \text{m/km}$$

	$\hat{H}$ (m)	σ (mm)
B	12.3487	1.55
C	9.8670	1.58
D	11.1220	1.42

$\hat{\sigma}_0 = 1.03$; $\det N = 0.1521$.

A3 · Error propagation in leveling

Random and systematic

$$\sigma_{\Delta H}^2 = \sum_i \sigma_i^2 = \eta^2 \sum_i K_i = \eta^2 K \;\Rightarrow\; \sigma = \eta\sqrt{K}$$

$$e = \delta K$$

$$\mathrm{RMSE}^2 = \eta^2 K + (\delta K)^2, \qquad K^* = (\eta/\delta)^2$$

With $\eta = 1\,\mathrm{mm}/\sqrt{\mathrm{km}}$ and $\delta = 0.1\,\mathrm{mm/km}$: $K^* = 100\,\mathrm{km}$; at 2000 km, 44.7 mm versus 200 mm.

Per-sight effects

$$c = (1-k)\frac{s^2}{2R}, \quad k \approx 0.13$$

$$e_{\mathrm{col}} = \varepsilon\, s, \qquad \Delta e = \varepsilon\,(s_{\mathrm{ré}} - s_{\mathrm{vante}})$$

$$w \approx H\,\frac{\Delta g}{\gamma} \quad \text{(loop)}$$

$s = 50$ m: $c = 0.20$ mm (0.17 with refraction); $\varepsilon = 10''$ and $\Delta s = 20$ m: 0.97 mm per setup; $H = 1000$ m, $\Delta g = 20$ mGal: $w = 20.4$ mm.

Forward and back: $d = \Delta h_{\mathrm{forward}} + \Delta h_{\mathrm{back}}$ (with the return of opposite sign); $\mathrm{Var}(d) = 2\sigma_1^2$ if each run has $\sigma_1 = \eta_1\sqrt{K}$. Section tolerance: $|d| \le 3\sqrt{K}$ mm; maximum standard error of an adjusted line: $2\sqrt{K}$ mm (IBGE, 1983, Table VI, p. 30).

A4 · Timeline of Brazilian leveling (1)

Year	Milestone
1919–20	**Torres** (RS) tide gauge, of the Comissão da Carta Geral: MSL from Feb 1919 to Feb 1920, provisional datum.
1945	October: the CNG (IBGE) starts precision leveling (Leveling Section; origin at Urussanga, SC).
1946	December: the network is tied to the Torres gauge.
1948	Imbituba gauge series begins (PSMSL: 1948–68). First adjustment (Torres datum).
1949–57	MSL observation window of **Imbituba** (9 years): the vertical datum.
1952	Second adjustment (Torres datum).
1957–58	MSL observation window of **Santana** (2 years).
1958 / 1959	Replacement in 1958 (IBGE, 2011, p. 8); adoption and third adjustment in 1959 (IBGE, 2019, p. 13). Adjustments use Imbituba from 1959 onward.
1980–81	Santana adopted in 1980; benchmarks established in 1980–81, using 1957–58 MSL (IBGE, 2019, pp. 13 and 44).
1991–92	Macro-loops and the AAGP (preliminary global adjustment); last partitioned adjustment: 43 079 benchmarks, with BDG calculation reference 1 December 1992 (IBGE, 2011, p. 12).

A4 · Timeline of Brazilian leveling (2)

Year	Milestone
1996	The RMPG idea is proposed by UFRJ and UFPR.
1997	SIRGAS WG III (Vertical Datum) is established during the IAG Scientific Assembly in Rio de Janeiro.
2000–01	SIRGAS2000 campaign with GPS at tide gauges; WG III consolidated in Cartagena (2001); the Geodesy Coordination begins to run the RMPG (2001).
2005–11	DALTI: simultaneous adjustment of the network with GHOST; released in June 2011 (69 590 benchmarks, $>$ 74 000 observations).
2010	IBGE Technical Manual No. 11: geodetic control of tide gauge stations (CGEM).
2015	IAG Resolution No. 1: defines the IHRS ($W_0 = 62\,636\,853.4\,\mathrm{m}^2/\mathrm{s}^2$); the regional realization should densify the IHRF.
2018–19	30/07/2018: REALT-2018 (geopotential numbers, real gravity, normal heights). 30/09/2019: 2nd edition of the report.
2020	hgeoHNOR2020: converts h (GNSS) to H^* (quasigeoid class).
2022	1 April: the RMPG also begins to publish meteorological data.
2024	Consolidated RMPG series: Jul 2001 to Dec 2024 (23.5 years).
2026	Example time reference: 72.5 years since 1953.5, the midpoint of Imbituba's MSL window. Trend extrapolation is illustrative.

A5 · The computation of the nodal cycle and mean level

Tides and nodal cycle

$$T_{M2} = 12.4206\,\mathrm{h}, \quad T_{S2} = 12.0000\,\mathrm{h}$$

$$T_{\text{viva-morta}} = \left(\frac{1}{T_{S2}} - \frac{1}{T_{M2}}\right)^{-1} = 14.77\,\mathrm{d}$$

$$T_{\text{nodal}} = 6798.38\,\mathrm{d} = 18.61\,\mathrm{yr}$$

$$N_{M2} = \frac{6798.38 \times 24}{12.4206} = 13\,136$$

$$\delta_{\max} = \varepsilon \pm i = 23.44^\circ \pm 5.15^\circ = 28.6^\circ,\ 18.3^\circ$$

How long to average? (simulation)

Synthetic model: annual cycle 8 cm, nodal cycle 2 cm, AR(1) noise $\sigma = 6$ cm, $\rho = 0.5$; 4000 series, fixed seed. Rms error of the estimated MSL:

1 month	8.3 cm
1 year	3.1 cm
2 years	2.5 cm
9 years	1.4 cm
19 years	0.7 cm

Illustrative parameters, not measured.

Sea rise: 2.24 $\mathrm{mm/yr} \times 72.5\ \mathrm{yr} = 162$ mm (± 29 mm from the rate of ± 0.4 mm/yr). Imbituba window: 1949–57, middle at 1953.5; to 2026: 72.5 years.

A6 · Bilingual glossary (1)

- **altitude / height**: vertical distance counted from the datum to the point (the "ruler").
- **altitude normal / normal height** (H^*): $C/\bar{\gamma}$.
- **altitude ortométrica / orthometric height** (H): $C/\bar{g}$.
- **número geopotencial / geopotential number** (C): $W_0 - W_P$, $\mathrm{m}^2/\mathrm{s}^2$.
- **datum vertical / vertical datum**: one MSL value, at one station and epoch, taken as zero altitude.
- **nível médio do mar (NMM) / mean sea level (MSL)**: average of the water level over a period.
- **marégrafo / tide gauge**: instrument that records the sea level.
- **sizígia / spring tide; quadratura / neap tide**.
- **ciclo nodal / nodal cycle**: 18.6 years, regression of the Moon's nodes.
- **nível de redução, zero hidrográfico / reduction level, chart datum**: reference plane of nautical charts.
- **nivelamento geométrico / geometric (spirit) leveling**: measurement of height differences with level and rod.
- **nível / level; mira / (leveling) rod**.
- **ré / backsight; vante / foresight**: Δh = back − fore.
- **lance / setup; seção / section; linha / line**.
- **circuito / loop; macrocircuito / macro-loop**.
- **fechamento / misclosure** (w): sum of the height differences of a loop.
- **tolerância / tolerance**: limit above which the section is redone.
- **referência de nível (RN) / benchmark (BM)**.

A6 · Bilingual glossary (2)

- **ajustamento / adjustment**: computation that reconciles redundant measurements.
- **mínimos quadrados / least squares**: minimizes $\sum p\, v^2$.
- **peso / weight** ($p = 1/(\eta^2 K)$); **resíduo / residual** (v).
- **injunção / constraint**: point (or value) held fixed in the adjustment.
- **rede livre / free network; rede injuncionada / constrained network**.
- **defeito de posto / rank defect**: lack of a datum.
- **redundância / redundancy** ($r = m - n$).
- **fator de variância / variance factor** ($\hat{\sigma}_0^2$).
- **erro grosseiro / blunder**; ***data snooping***: test of standardized residuals.
- **erro aleatório / random error; sistemático / systematic**.
- **colimação / collimation; refração / refraction; curvatura / curvature**.
- **gravidade / gravity; gravimetria / gravimetry**.
- **quase-geoide / quasigeoid** (ζ); **geoide / geoid** (N).
- **GNSS**: global navigation satellite systems.
- **subsidência / subsidence**: sinking of the ground.
- **época / epoch**: date to which a value refers.

A7 · Acronyms and institutions (1)

Acronym	Name and meaning
CNG	National Geography Council (an IBGE body; started precision leveling in 1945).
IBGE	Brazilian Institute of Geography and Statistics (the country's geodetic authority).
SGB	Brazilian Geodetic System: planimetric, vertical and gravimetric networks.
RAAP	High-Precision Altimetric Network: the SGB leveling network.
RN, RRNN	Benchmark, benchmarks.
REALT	Readjustment of the Altimetric Network with Geopotential Numbers (REALT-2018; 2nd ed. 2019).
AAGP	Preliminary Global Altimetric Adjustment (1991–92).
DALTI	Altimetry team/project of IBGE's Geodesy Coordination (2011 simultaneous adjustment).
GHOST	*Geodetic adjustment using Helmert blocking Of Space and Terrestrial data* (Canadian software).
BDG	IBGE Geodetic Database (station sheets).

A7 · Acronyms and institutions (2)

Acronym	Name and meaning
RMPG	Permanent Tide Gauge Network for Geodesy (IBGE; six stations in 2024).
CGEM	Geodetic Control of Tide Gauge Stations (IBGE Technical Manual No. 11, 2010).
DHN	Directorate of Hydrography and Navigation (Brazilian Navy; nautical charts and hydrographic zeros).
PSMSL	*Permanent Service for Mean Sea Level* (sea level series, e.g., Imbituba 1948–68).
NOAA / NTDE	*National Oceanic and Atmospheric Administration* / *National Tidal Datum Epoch* (19-year datum epoch).
IAG	*International Association of Geodesy*.
IHRS / IHRF	*International Height Reference System* / *Frame* (IAG, 2015; $W_0 = 62\,636\,853.4\,\mathrm{m}^2/\mathrm{s}^2$).
SIRGAS, GT III	Geocentric Reference System for the Americas; Working Group III (Vertical Datum).
RBMC	Brazilian Network for Continuous Monitoring of GNSS (e.g., station IMBI at Imbituba).
BNDG	National Gravity Database.
MAPGEO2015, hgeoHNOR2020	IBGE geoid model; conversion model $h \to H^*$ (quasigeoid class).

A8 · Primary sources (1)

- **[K]** IBGE (1983). *Especificações e Normas Gerais para Levantamentos Geodésicos em Território Brasileiro*. R.PR 22, 21/07/1983, Table VI, p. 30 (PDF p. 26): field checks and standard error after adjustment.
- **[B]** IBGE (2011). *Ajustamento Simultâneo da Rede Altimétrica de Alta Precisão do Sistema Geodésico Brasileiro*. p. 2: 69 590 benchmarks; p. 8: origins and datum replacement in 1958; pp. 10–12: eight adjustments, macro-loops and AAGP, reference 1 December 1992.
- **[C]** IBGE (2019). *Reajustamento da Rede Altimétrica com Números Geopotenciais*, 2nd ed. ISBN 978-85-240-4502-8. p. 13: adoption of Imbituba in 1959 and Santana in 1980; p. 35: principal block; p. 36, Figure 16: residual frequencies; pp. 40–44: individual lines and Amapá.

A8 · Primary sources (2) and further reading

- **[D]** Siqueira, J.; Lima, M.; Luz, R. (2018). *Reajustamento da Rede Altimétrica Brasileira em Números Geopotenciais*. Simposio SIRGAS2018, Boletín 23. slides 3–4: adjustment dates; 9: methodology change; 20–21: interpolation; 31–36: results.
- **[I]** SIRGAS, *Working Group III (Vertical datum)*. Established at the 1997 IAG Scientific Assembly; IHRS-compatible definition; realization as IHRF densification, with a reference epoch.
- **[A, E]** IBGE, *Rede Altimétrica*; *Controle Geodésico de Estações Maregráficas* (2010); RMPG reports. [J] NOAA, *Tidal Datums*: 19-year epoch.
- **[G, H]** MSL/subsidence rates and the Imbituba/Santana offset are study estimates (2013–2019), not official transformation parameters. References and verification limits are in `sgb_altimetria_fontes.md`.

A8 · Different quantities in quality assessment

Quantity	Value	Meaning
Observation noise	$\sigma = \eta\sqrt{K}$	Stochastic model; η selected or estimated
Section forward/backward discrepancy	$3\,\mathrm{mm}\sqrt{K}$	Field check, item 3.1
Line forward/backward discrepancy	$4\,\mathrm{mm}\sqrt{K}$	Field check, item 3.2
Circuit misclosure	$\lvert w\rvert/K \le 0.5\,\mathrm{mm/km}$	After corrections, item 3.3
Adjusted-line standard error	$2\,\mathrm{mm}\sqrt{K}$	Maximum acceptable after adjustment, item 4
REALT-2018 residuals	92.7–99.5 %	Empirical principal-block frequencies

IBGE (1983), Table VI, p. 30, fundamental high precision; IBGE (2019), p. 36. K: length in km, counted once. Normalization by $v/\sqrt{K}$ depends on distance. The statistical test v_i/σ_{v_i} in Appendix A2 is dimensionless and serves a different purpose.

A8 · Scope of the figures and qualifications

- **Torres/Imbituba**: IBGE (2011), p. 8, gives 1958; IBGE (2019), p. 13, gives adoption in 1959. Whether these refer to the same historical act remains unresolved.
- **Counts**: 66 882 benchmarks describes the REALT-2018 principal block. A national total is not inferred from that table. The 69 590 benchmarks and more than 74 000 observations belong to the 2011 report.
- **Estimates**: datum offsets, MSL trend and subsidence retain their study qualifications. Projection to 2026 assumes a constant rate.
- **Illustrative examples**: synthetic series, fictitious benchmarks, four-node network, GNSS residuals and 230 000/370 000 km lengths. Schematic drawings are not maps.